COURSE SLIDES · WSQ

Python Programming for Finance

Course Code: TGS-2026065049

3 days · 24 hours · Hands-on financial analytics
Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W



Version v1.4 · 24 August 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

3 days · 24 hours · finance-focused Python practice

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Digital Attendance (Mandatory)

1

AM attendance

Scan the TRAQOM/SSG QR code at the start of each day.

2

PM attendance

Complete the second attendance window when directed.

3

Assessment attendance

Take the separate digital attendance before the assessment.

4

Eligibility

At least 75% attendance and competency in all assessments are required.

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About the Trainer



Your Trainer

General Trainer template

NAME _____

TITLE / DESIGNATION _____

QUALIFICATIONS _____

AREAS OF EXPERTISE _____

TRAINING EXPERIENCE _____

CONTACT _____

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE	Founder and Managing Director
QUALIFICATIONS	PhD, MEng and MBA
EXPERIENCE	25+ years in IT, AI and financial systems
DELIVERS	Accredited data, AI and programming programmes
FOCUS	Applied analytics and accountable automation

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ICE-BREAKER

Let’s Know Each Other

1

You

Name, organisation and role

2

Experience

Python, Excel or data analysis experience

3

Finance task

A report or decision you want to improve

4

Success

What a useful result looks like after three days

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HOUSEKEEPING

Ground Rules

1

Participate

Ask questions and share relevant experience.

2

Respect

One conversation at a time; disagree constructively.

3

Protect data

Use only supplied or authorised finance data.

4

Be punctual

Return from breaks and lunch on time.

5

Validate

Do not trust output only because it looks plausible.

6

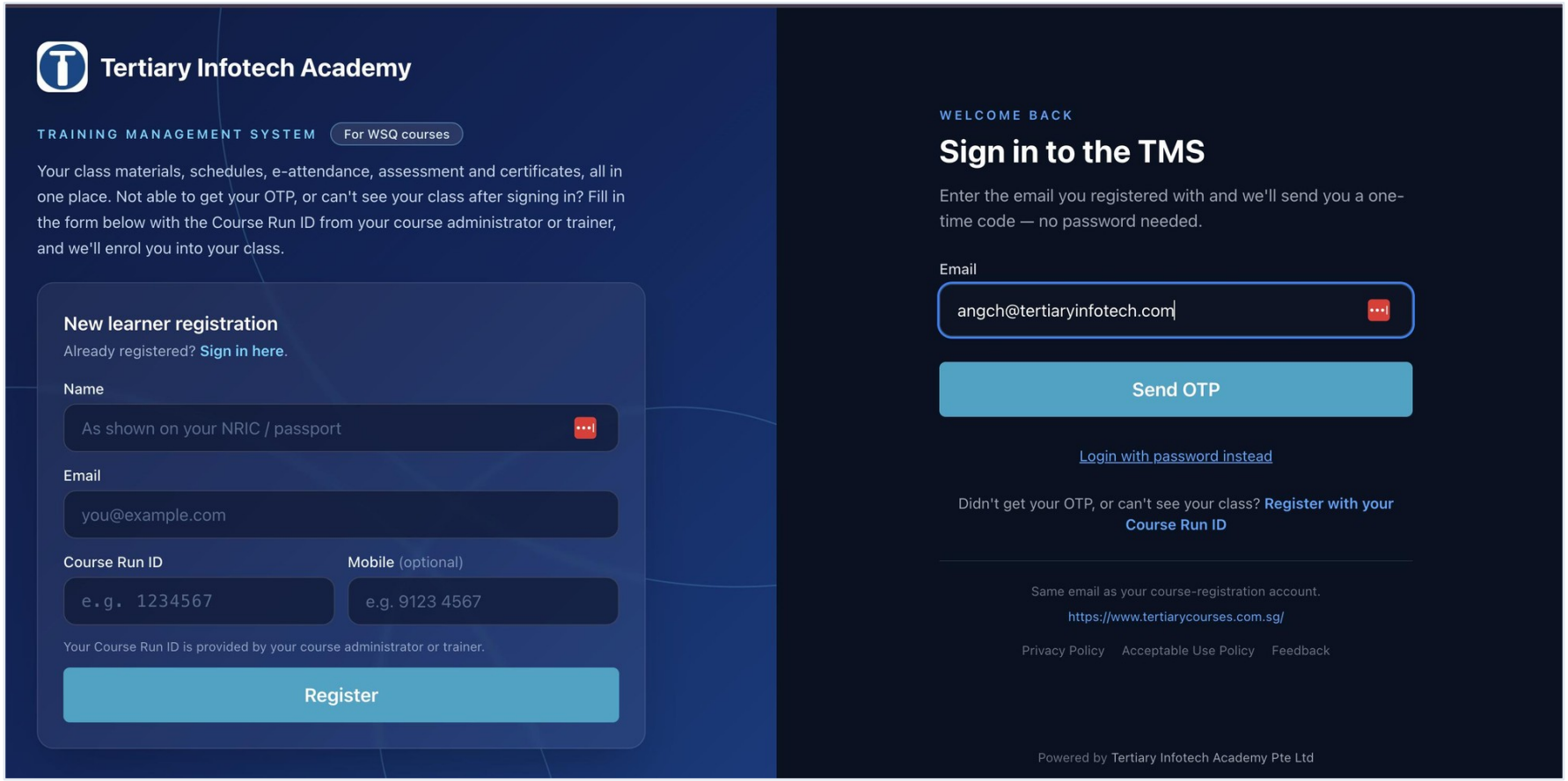
Support

Help peers explain decisions, not copy answers.

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COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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COURSE PORTAL · LMS/TMS

Download Course Material

https://lms-tms.tertiaryinfotech.com/

1 · SIGN IN — Use the course invitation or trainer-provided credentials.

2 · OPEN COURSEWARE — Choose the current Python Programming for Finance course.

3 · DOWNLOAD — Save the slides, Learner Guide and lab folders locally.

4 · ASSESS — Open and submit the learner-facing papers only when directed.

COURSE PACKAGE

- Visual course slides
- Detailed Learner Guide
- 17 self-contained lab folders
- Starter code and offline data
- Assessment papers on the LMS

Open LMS/TMS

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SCHEDULE

Lesson Plan — 3 Days, 24 Hours

DAY 1

Python foundations and finance logic

- Topics 1–3
- Labs 1–6
- Python and finance logic

9:00 AM – 6:00 PM

DAY 2

Reusable code and Pandas data preparation

- Topics 4–5
- Labs 7–11
- Functions and data preparation

9:00 AM – 6:00 PM

DAY 3

Aggregation, visualisation and financial analysis

- Topics 6–7
- Labs 12–17
- 2-hour final assessment

9:00 AM – 6:00 PM

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WHAT YOU WILL ACHIEVE

Learning Outcomes

1

LO1

Translate finance business requirements into clear Python programming objectives and configure a reproducible development environment.

2

LO2

Use Python data types and operators to represent, calculate and present financial information accurately.

3

LO3

Organise finance logic with conditional statements, loops and comprehensions to solve repeatable analytical problems.

4

LO4

Create reusable functions and lambda-based transformations for finance business use cases.

5

LO5

Import, inspect, filter, slice and clean finance data with Pandas Series and DataFrames.

6

LO6

Join, aggregate, test and visualise finance data using groupby, pivot tables and appropriate charts.

7

LO7

Improve analytical code with apply and pipe, use descriptive statistics, and analyse changes in financial data over time.

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BEFORE YOU START

Briefing for Assessment

1

Individual

Complete the papers independently.

2

Open book

Use approved course materials only.

3

Evidence

Show calculations, code and requested outputs.

4

Conduct

No discussion, photos or recording.

5

Submission

Upload completed question papers on the LMS.

6

Sign-off

Complete the Assessment Summary Record.

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FINAL ASSESSMENT

Assessment

1

WA · 60 min

8 open-ended short-answer questions covering K1–K8.

2

PP · 60 min

5 practical tasks covering A1–A9 and aligned to Labs 1, 13 and 14.

3

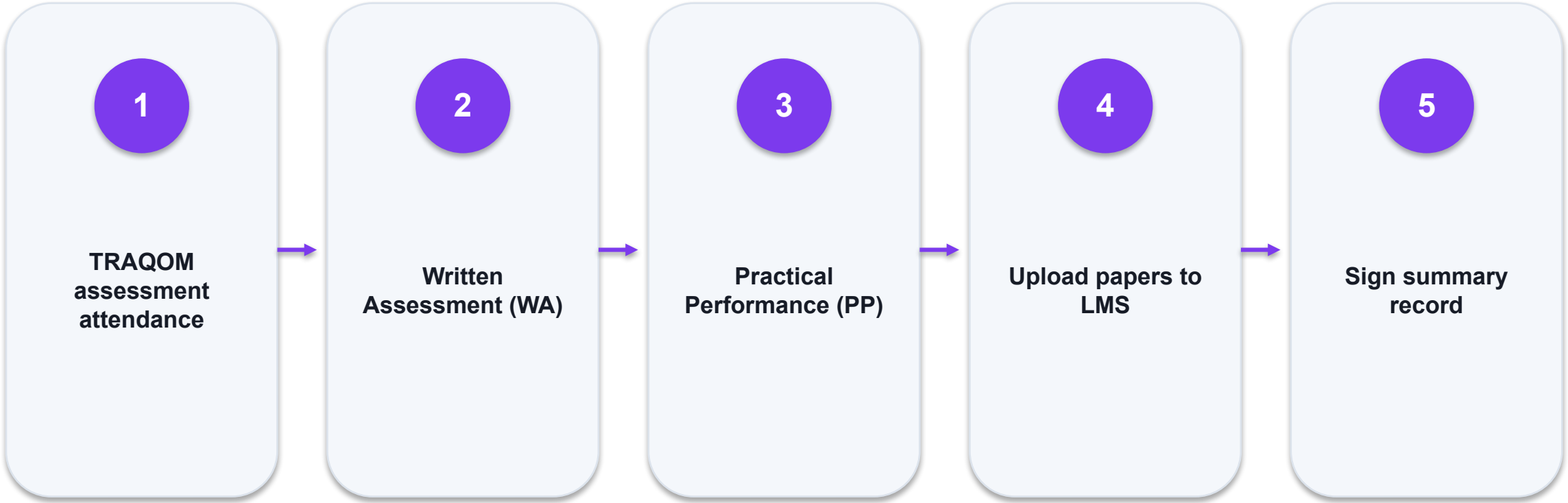
Competency

Both instruments are open book; demonstrate correct, explainable work.

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ON ASSESSMENT DAY

Assessment Flow



The static slide contains the full sequence; trainers may reveal stages progressively.

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LEARNER RESOURCES

Access the Hands-On Labs

https://github.com/tertiarycourses/TGS-2026065049-Python-Programming-for-Finance

Each lab folder is self-contained: README, PDF guide, starter.py, solution.py, tests and the data needed to run offline.

Detailed click-by-click procedures are in the Learner Guide and lab README—not in the concept-led slide deck.

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TWO WAYS TO GET THE PUBLIC LABS

Access the Hands-On Labs

Option A · Clone

git clone
https://github.com/tertiarycourses/TGS-2026065049-Python-Programming-for-Finance.git

Keeps the folder structure and makes updates easy.

Option B · Download ZIP

GitHub → Code → Download ZIP

Extract it locally before opening a lab folder.

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CORE CONCEPTS

Python, Data and Decisions in Finance

A controlled analytical workflow from requirement to evidence

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FINANCE USE CASES

Where Python Creates Value in Finance



1 **Market + portfolio** — Acquire prices; calculate value, weights, returns and risk.

2 **Credit decisions** — Encode affordability, segmentation and review thresholds.

3 **Reporting + controls** — Automate tables and charts; reconcile inputs and exceptions.

4 **Rapid prototyping** — Turn an analytical idea into tested, reviewable Python logic.

DECISION / CONTROL

Python speeds up the analytical chain; validated data, reconciliations and accountable review make the result decision-ready.

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Generative AI vs Agentic AI: Different Units of Work

COMPARE	GENERATIVE AI	AGENTIC AI
Unit of work	A prompt plus supplied context.	A goal plus constraints and success criteria.
Run pattern	One or a few model responses produce a candidate output.	Plan → use tools → observe → evaluate; repeat until done or stopped.
Tools and action	May suggest code or analysis; a person usually executes it.	Invokes permitted Python functions, data sources, APIs or workflow actions.
State	Often limited to the request or conversation context.	Maintains task state and checkpoints across multiple steps when engineered.
Completion	A draft, summary, explanation or code suggestion is ready for review.	The objective passes checks, or an exception is routed with evidence.
Human responsibility	Validate the content, then decide and act.	Set permissions and policies; approve material actions and exceptions.
Primary risk	Plausible but unsupported content or code.	An error or unauthorised action can propagate through connected tools.

Finance example · Generative AI drafts variance commentary from supplied tables. Agentic AI can retrieve approved data, run Python checks, investigate exceptions, attach evidence and route the pack for approval.

Source: OpenAI Agents SDK · Agents and agent loop

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The Value of Agentic AI Is Controlled Workflow Closure

MONTHLY VARIANCE PACK · EXECUTION TRACE

1. **GOAL + POLICY** — Define the reporting period, materiality rule and allowed data sources.
2. **RETRIEVE** — Read the approved general-ledger and budget extracts.
3. **RECONCILE IN PYTHON** — Match entities and accounts; stop if totals or row counts fail.
4. **INVESTIGATE** — Calculate material variances and attach source evidence.
5. **APPROVE + PUBLISH** — Route the draft to an accountable reviewer before release.

STOP RULE · No publication when reconciliation fails or approval is missing.

VALUE LEVER · MEASURE IT

- **CYCLE TIME** — Measure end-to-end elapsed time, including waits and retries.
- **COVERAGE** — Measure the percentage of entities and records checked each run.
- **CONTROL CONSISTENCY** — Measure runs completing every mandatory reconciliation.
- **HUMAN LEVERAGE** — Measure analyst time redirected to exceptions and judgement.

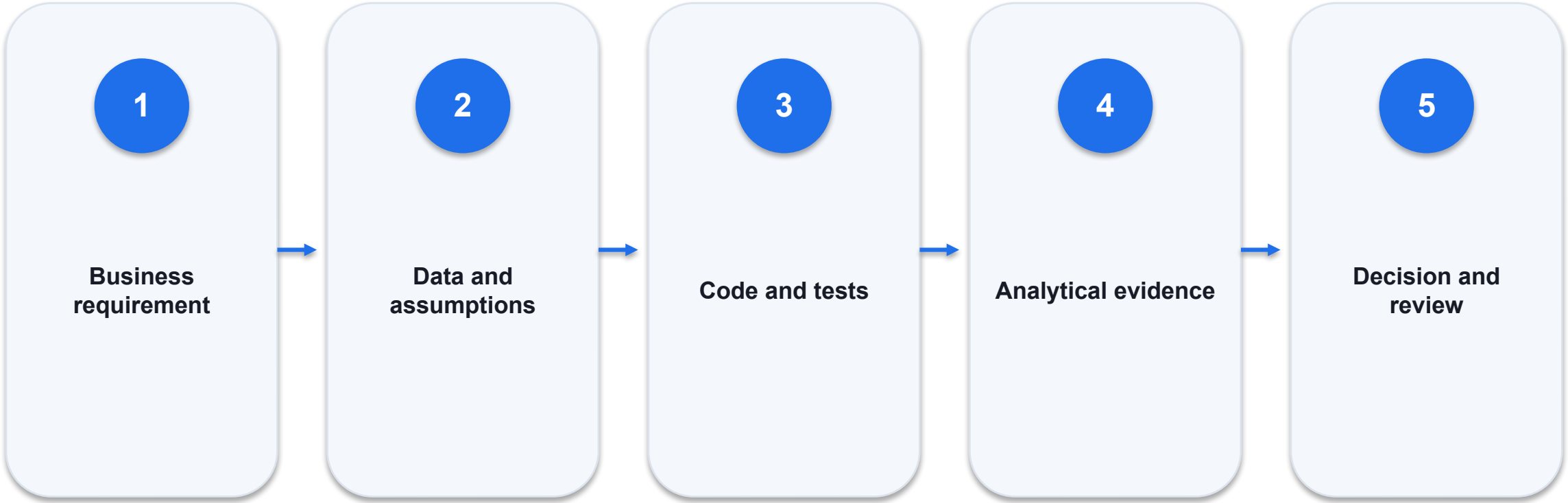
Value is valid only inside permissions, checks and human accountability.

Sources: OpenAI Agents SDK; NIST AI RMF Playbook

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END-TO-END METHOD

The Controlled Finance Analytics Lifecycle



The static slide contains the full sequence; trainers may reveal stages progressively.

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CONTROL FRAMEWORK

The Four Controls on Every Result

1

Completeness

Did all expected rows, periods and instruments arrive?

2

Validity

Are types, ranges and categories defensible?

3

Accuracy

Do known examples and reconciliations match?

4

Explainability

Can another person trace data, assumptions and formulas?

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01

START OF DAY 1

Day 1

Python foundations and finance logic

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DAY 1 · ORIENTATION

Day 1 Learning Route

1

Reconnect

Review the previous day’s evidence and open questions.

2

Concepts

Build the mental model before touching the code.

3

Application

Use finance scenarios and offline-safe data.

4

Evidence

Reconcile results and explain what they mean.

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TOPIC 01

Introduction to Python Programming

Business requirements · finance use cases · environment · first script

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LO1 · CONCEPT MAP

Key Concepts — Introduction to Python Programming

1

Business first

Define the decision, users, data, output and acceptance criteria before writing code.

2

Finance fit

Python connects market data, calculations, visualisation, automation and AI in one ecosystem.

3

Reproducibility

A locked environment makes analytical results easier to review, repeat and audit.

4

Accountability

The analyst remains responsible for validating inputs, formulas, assumptions and outputs.

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Why Python Is Used in Finance



1

Analysis

Pandas and NumPy turn tables and time series into repeatable calculations.

2

Automation

Scripts remove manual copy-and-paste from recurring reports.

3

Decision support

Charts, statistics and rules surface trends, risk and exceptions.

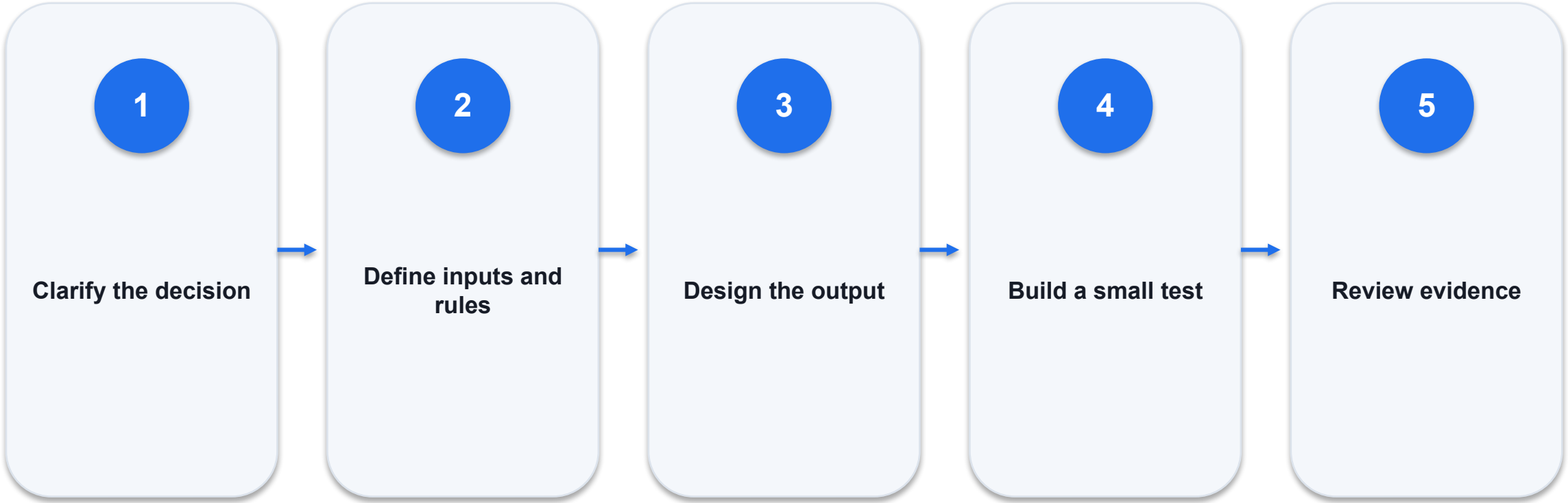
DECISION / CONTROL

Python creates value when a finance requirement becomes transparent code, testable calculations and reviewable evidence.

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TOPIC 01 · PROCESS

From a business requirement to working code



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TOPIC 01 · COMPARISON

Finance coding risk: plausible is not correct

Looks plausible	Is validated
A chart renders and totals appear reasonable.	Inputs reconcile, formulas are tested, edge cases are handled and results are explainable.

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TOPIC 01 · DECISION FRAME

Choosing a Python workspace

1

Google Colab

Fast browser start; easy sharing; cloud session.

2

VS Code

Local files, extensions, debugger and source control.

3

JupyterLab

Exploration with narrative, code and charts together.

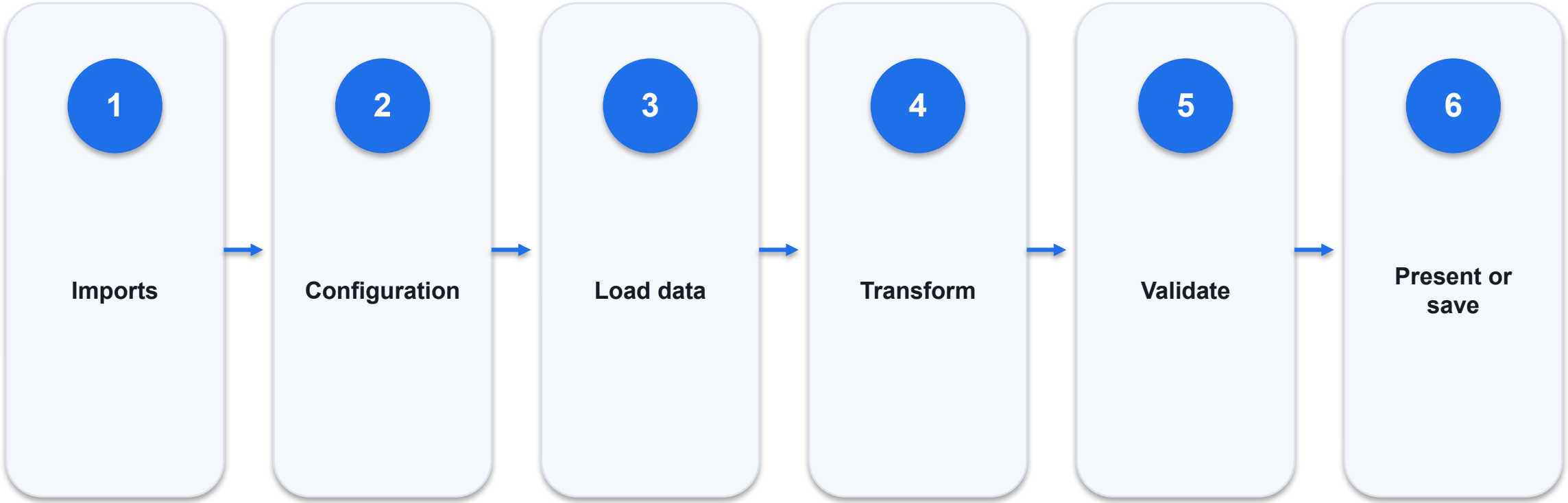
4

uv project

Fast dependency management and reproducible environments.

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The anatomy of a finance script



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TOPIC 01 · CONCEPT

Acceptance criteria make results testable

1

Correctness

Known inputs produce expected outputs.

2

Resilience

Missing or invalid data is handled safely.

3

Usability

Output is clear, labelled and decision-ready.

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A requirement becomes a test

```
requirement = {'ticker':'D05.SI', 'currency':'SGD'}

accept = 'five prices, 2 decimals, no crash on empty data'
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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Guard the empty-data case

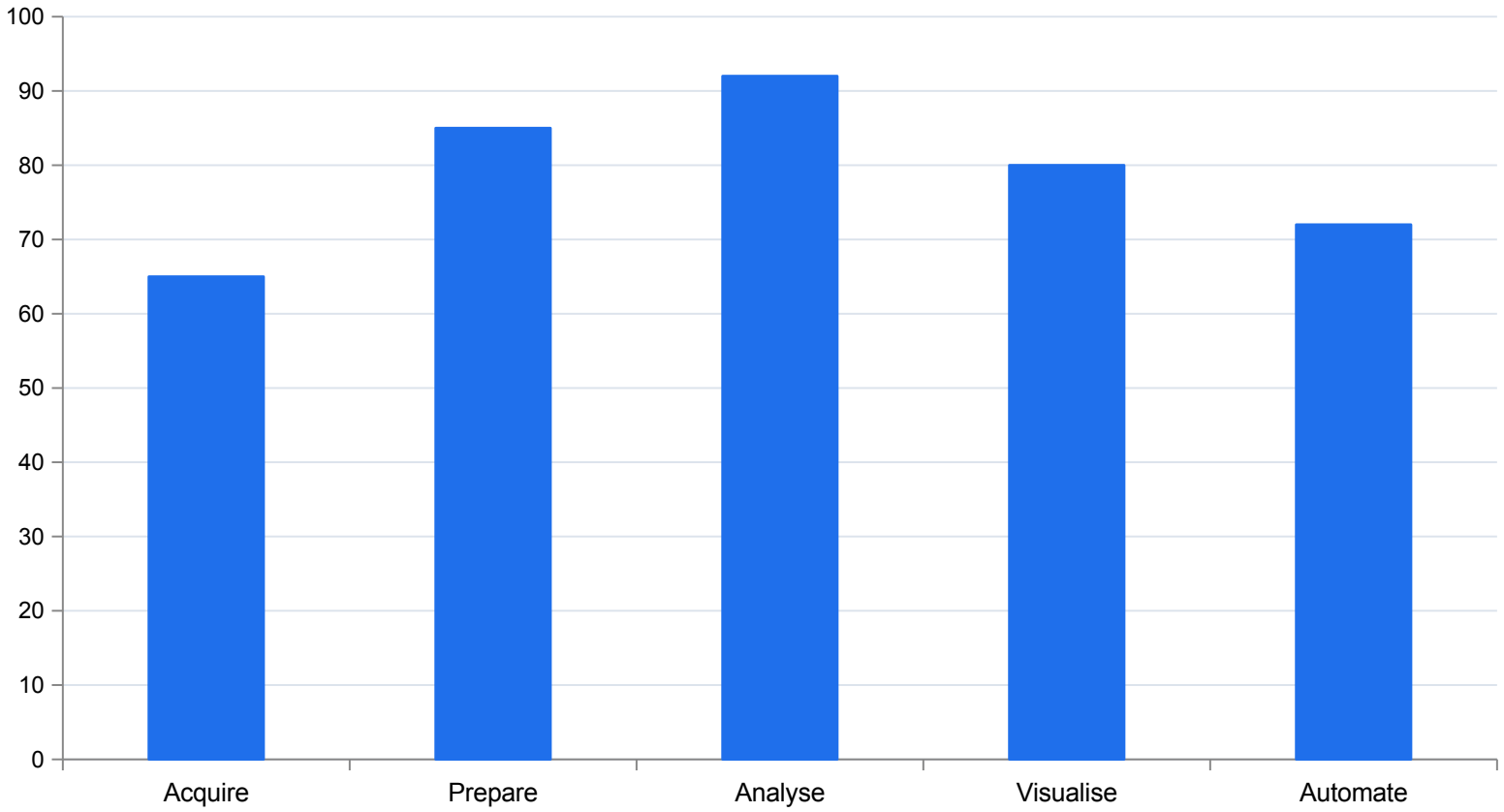
```
1 if prices.empty:
2     return {'status': 'NO DATA', 'rows': 0}
3 return {'status': 'OK', 'rows': len(prices)}
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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Python connects the finance workflow



WHAT THE DATA SHOWS

Python supports the full analytical chain; controls and validation sit across every stage.

Illustrative capability coverage (%)

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WHAT YOU WILL APPLY

Hands-On Labs — Introduction to Python Programming

1

Lab 01

Set Up a Reproducible Finance Project

2

Lab 02

Translate a Finance Requirement into an SGX Price Tool

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TOPIC 01 · HANDS-ON · LAB 01

Set Up a Reproducible Finance Project

FINANCE SCENARIO

A bank analytics team needs every learner to run the same package versions and verify the environment before analysing SGX data.

YOU WILL PRODUCE

A verified Python project with an environment check and dependency record.

TOOLS & EVIDENCE

Python 3, uv or venv, Pandas, Matplotlib

ALIGNMENT

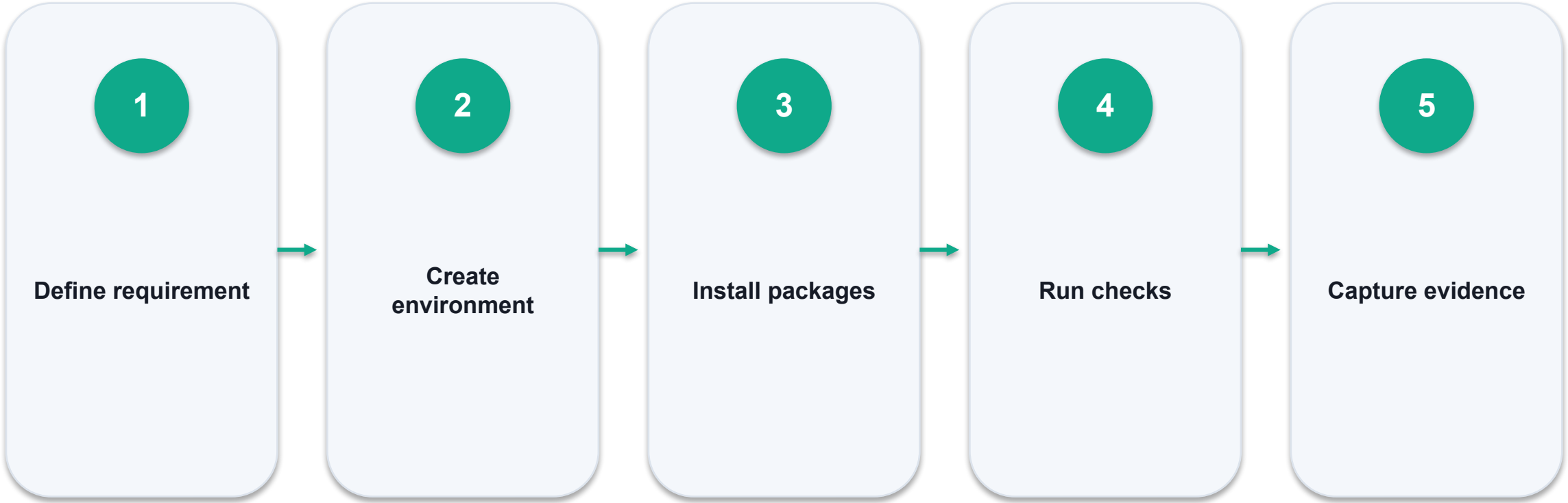
LO1 / A1

Folder: labs/lab-01-*

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LAB 01 · CONCEPT MAP (DETAILED PROCEDURE IN THE LEARNER GUIDE)

Set Up a Reproducible Finance Project — Analytical Workflow



The static slide contains the full sequence; trainers may reveal stages progressively.

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LAB 01 · VERIFY

Set Up a Reproducible Finance Project — Evidence & Acceptance

ACCEPTANCE TEST

The script prints the Python, Pandas, NumPy and Matplotlib versions and ends with ENVIRONMENT READY.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC 01 · HANDS-ON · LAB 02

Translate a Finance Requirement into an SGX Price Tool

FINANCE SCENARIO

A wealth desk wants a repeatable daily price summary for a small basket of Singapore-listed equities.

YOU WILL PRODUCE

A requirement canvas, acceptance tests and an offline-safe SGX price summary script.

TOOLS & EVIDENCE

Python, CSV, optional yfinance

ALIGNMENT

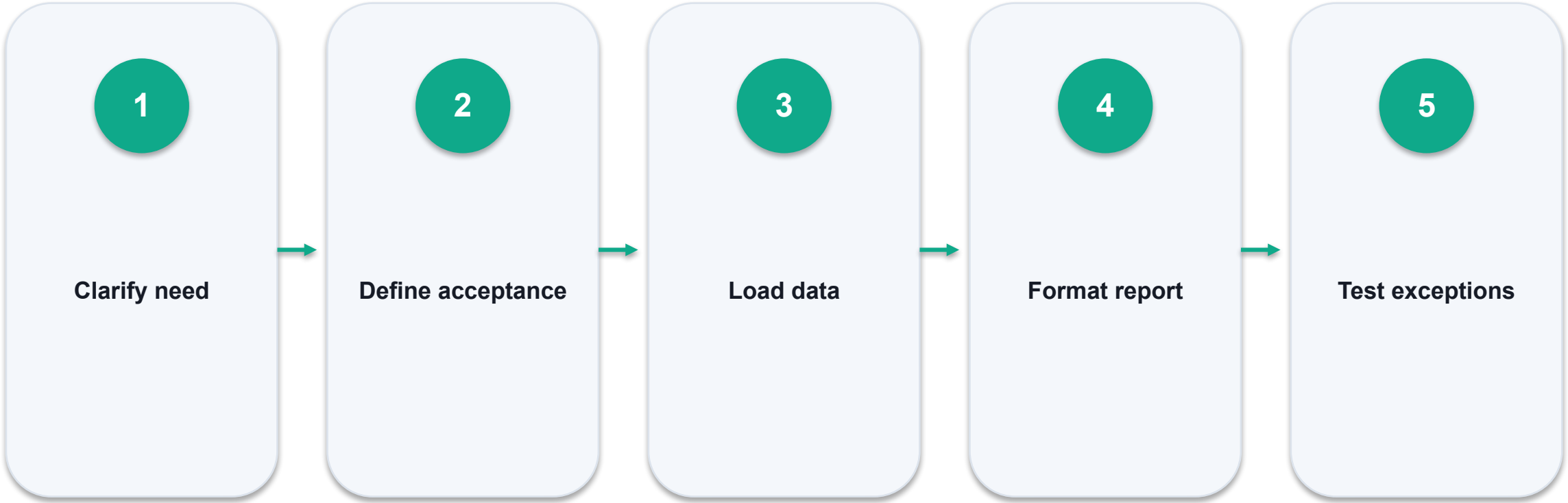
LO1

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LAB 02 · CONCEPT MAP (DETAILED PROCEDURE IN THE LEARNER GUIDE)

Translate a Finance Requirement into an SGX Price Tool — Analytical Workflow



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LAB 02 · VERIFY

Translate a Finance Requirement into an SGX Price Tool — Evidence & Acceptance

ACCEPTANCE TEST

The report prints five labelled SGX prices to two decimals, handles an invalid ticker without crashing, and all tests pass.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC RECAP

Recap — Introduction to Python Programming

1

Concept

Define the decision, users, data, output and acceptance criteria before writing code.

2

Method

Why Python Is Used in Finance

3

Application

Lab 1, Lab 2

4

Evidence

Validate outputs and explain the finance meaning.

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02

TOPIC 02

Data Types and Operators

Numbers · strings · lists · tuples · dictionaries · sets · operators

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LO2 · CONCEPT MAP

Key Concepts — Data Types and Operators

1

Numbers

Use int for counts and float or Decimal for measured financial values.

2

Strings

Clean labels, identifiers, dates and report text before matching or presenting them.

3

Collections

Lists, tuples, dictionaries and sets organise positions, identifiers and attributes.

4

Operators

Arithmetic, comparison and logical operators encode calculations and decisions.

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TOPIC 02 · COMPARISON

Financial numbers: precision and scale

float	Decimal
Fast and common for analytics; binary representation can create tiny rounding artefacts.	Base-10 arithmetic suited to currency rules where exact decimal behaviour matters.

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Formatting values for decision-makers

1

Currency

Thousands separators and two decimal places

2

Percentage

Multiply ratio by 100 and label the unit

3

Basis points

One basis point equals 0.01 percentage point

4

Large values

Use K, M or B only when the scale stays explicit

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Collection choice is a design decision

1. **list**

   Ordered, mutable sequence of prices or positions

2. **tuple**

   Fixed instrument identifier or multiple return values

3. **dict**

   Ticker-to-price or named financial attributes

4. **set**

   Unique holdings and portfolio overlap

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Operator families in finance

1

Arithmetic

+ - * / ** % for value, return, growth and allocation.

2

Comparison

Threshold rules such as ROE > 10 or PE <= 20.

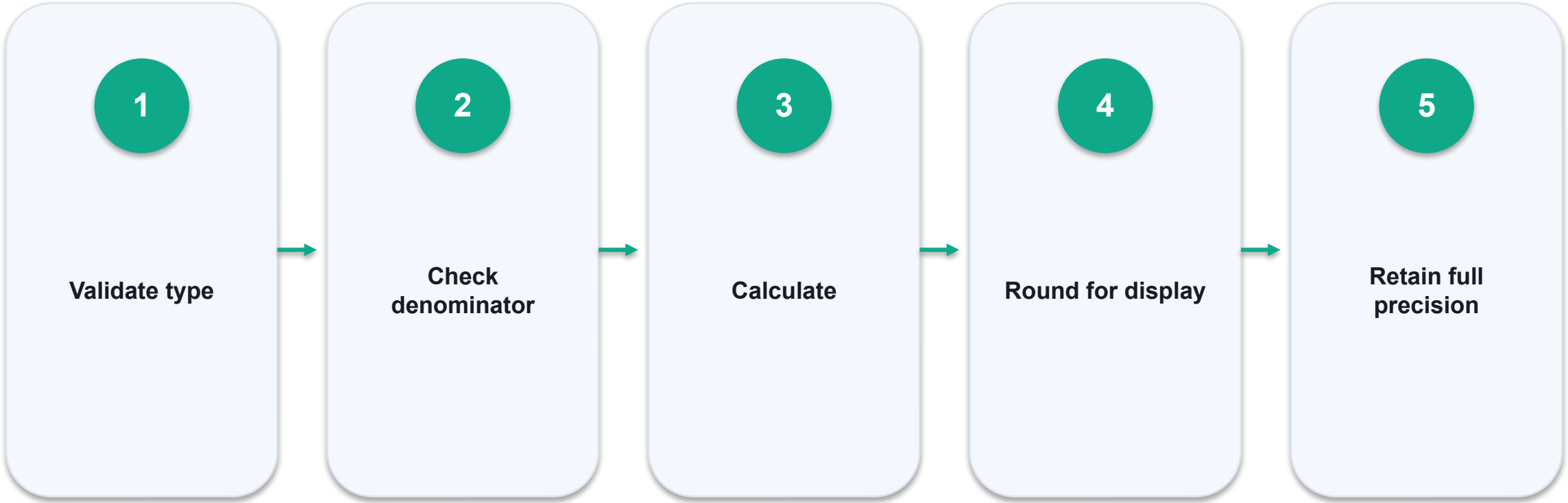
3

Logical

Combine conditions with and, or and not.

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Safe calculation patterns



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TOPIC 02 · COMPARISON

Data structures shape downstream analysis

Row-oriented records	Parallel lists
A list of dictionaries is readable and converts naturally to a DataFrame.	Easy to start, but alignment errors grow as fields are added.

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Currency formatting keeps units visible

```python
value = 12_487_650.75

print(f'SGD {value:,.2f}')  # SGD 12,487,650.75
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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A dictionary makes attributes explicit

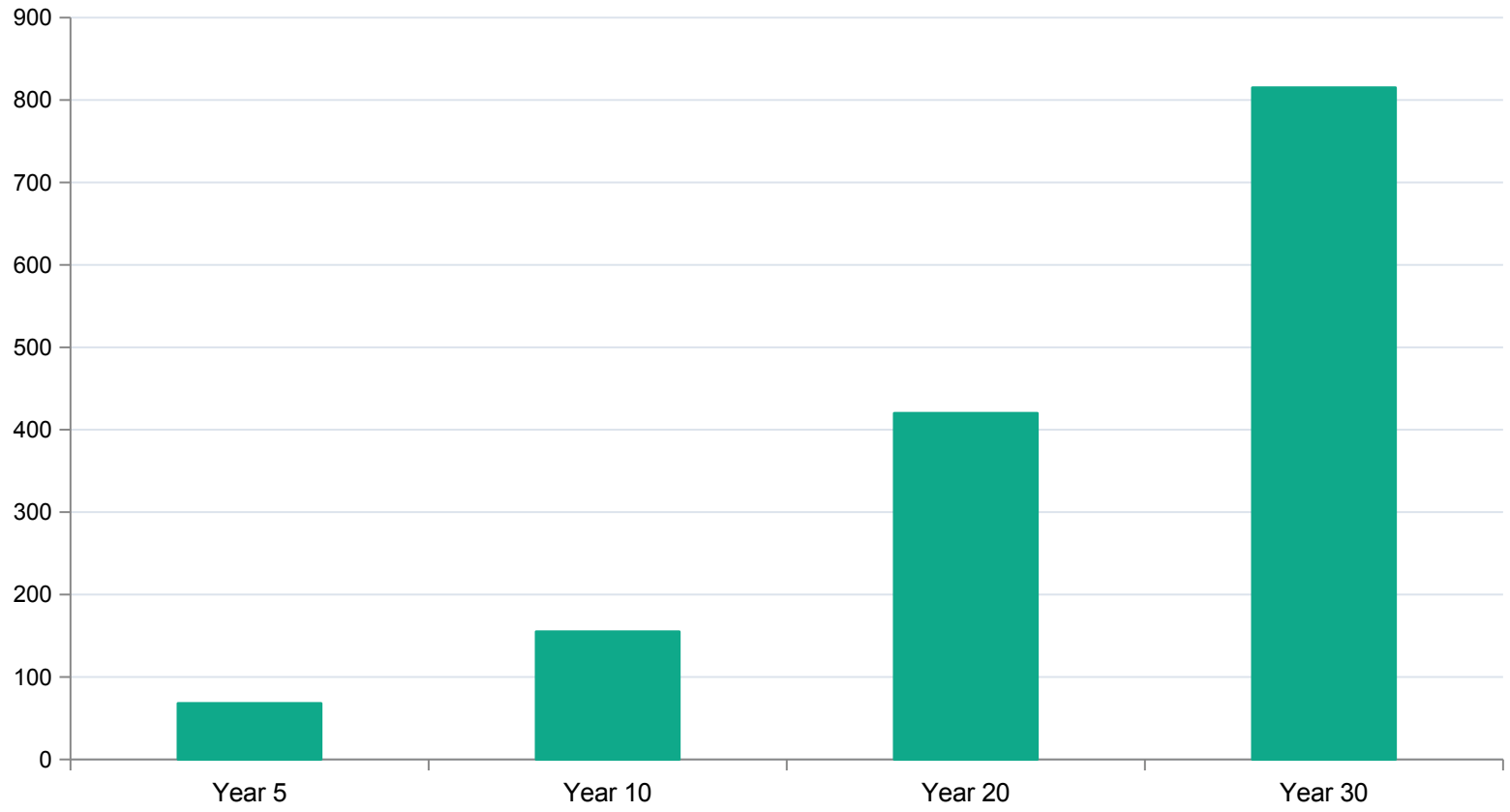
```
holding = {'ticker':'D05.SI', 'shares':120,
'price':38.42}
value = holding['shares'] * holding['price']
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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Compounding separates contribution from growth



WHAT THE DATA SHOWS

Compounding becomes dominant over longer horizons; assumptions should remain visible.

Illustrative savings balance (SGD '000)

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WHAT YOU WILL APPLY

Hands-On Labs — Data Types and Operators

1

Lab 03

Build a Loan Amortisation Calculator

2

Lab 04

Model a Portfolio with Python Collections

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TOPIC 02 · HANDS-ON · LAB 03

Build a Loan Amortisation Calculator



FINANCE SCENARIO

A relationship manager needs a transparent monthly-instalment calculation including the zero-interest staff-loan case.

YOU WILL PRODUCE

A loan calculator with monthly payment, total repayment and total interest.

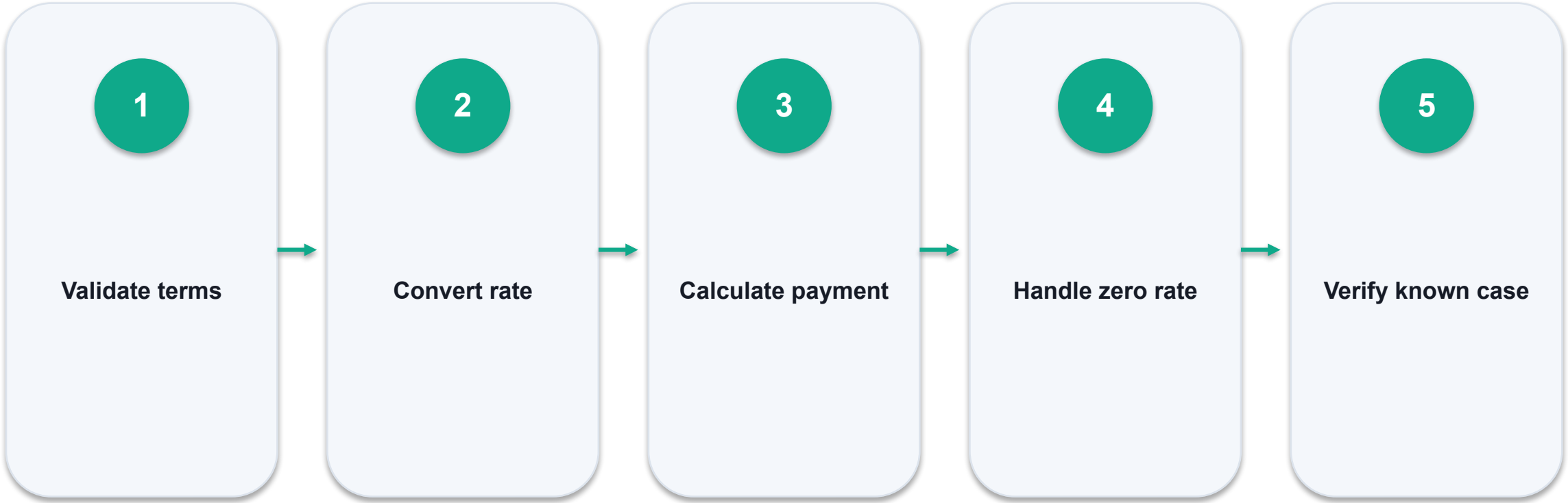
CONTROL **Reconcile principal, interest and ending balance; test zero-rate and boundary inputs.**

Python, math, csv · Folder: labs/lab-03-*

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LAB 03 · CONCEPT MAP (DETAILED PROCEDURE IN THE LEARNER GUIDE)

Build a Loan Amortisation Calculator — Analytical Workflow



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LAB 03 · VERIFY

Build a Loan Amortisation Calculator — Evidence & Acceptance

ACCEPTANCE TEST

For $500,000 at 2.6% over 25 years, payment is about $2,268.35; a 0% loan returns principal divided by months.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC 02 · HANDS-ON · LAB 04

Model a Portfolio with Python Collections

FINANCE SCENARIO

An investment team needs a clean in-memory representation of holdings, identifiers, sector groups and prices.

YOU WILL PRODUCE

A portfolio report using list, tuple, dictionary and set operations.

TOOLS & EVIDENCE

Python, csv

ALIGNMENT

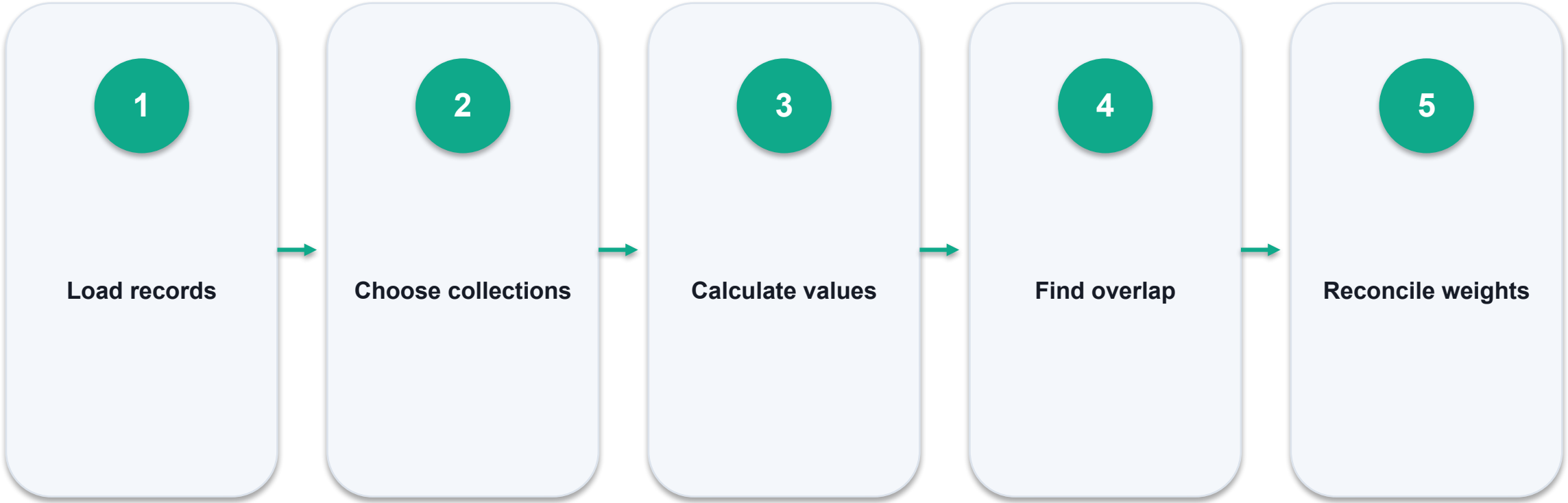
LO2

Folder: labs/lab-04-*

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LAB 04 · CONCEPT MAP (DETAILED PROCEDURE IN THE LEARNER GUIDE)

Model a Portfolio with Python Collections — Analytical Workflow



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LAB 04 · VERIFY

Model a Portfolio with Python Collections — Evidence & Acceptance

ACCEPTANCE TEST

Every holding has a market value, portfolio weights sum to 100%, and the overlap set is reported without duplicates.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC RECAP

Recap — Data Types and Operators

1

Concept

Use int for counts and float or Decimal for measured financial values.

2

Method

Financial numbers: precision and scale

3

Application

Lab 3, Lab 4

4

Evidence

Validate outputs and explain the finance meaning.

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03

TOPIC 03

Problem Solving with Control Structures

If/elif/else · loops · range · enumerate · zip · comprehensions

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LO3 · CONCEPT MAP

Key Concepts — Problem Solving with Control Structures

1

Branching

Use if/elif/else when a finance rule selects one outcome from alternatives.

2

Iteration

Use loops to apply the same logic to positions, periods or scenarios.

3

Loop control

break stops a search; continue skips an invalid record without ending the run.

4

Comprehensions

Create a transformed collection concisely when the rule remains readable.

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Turn policy thresholds into explicit branches



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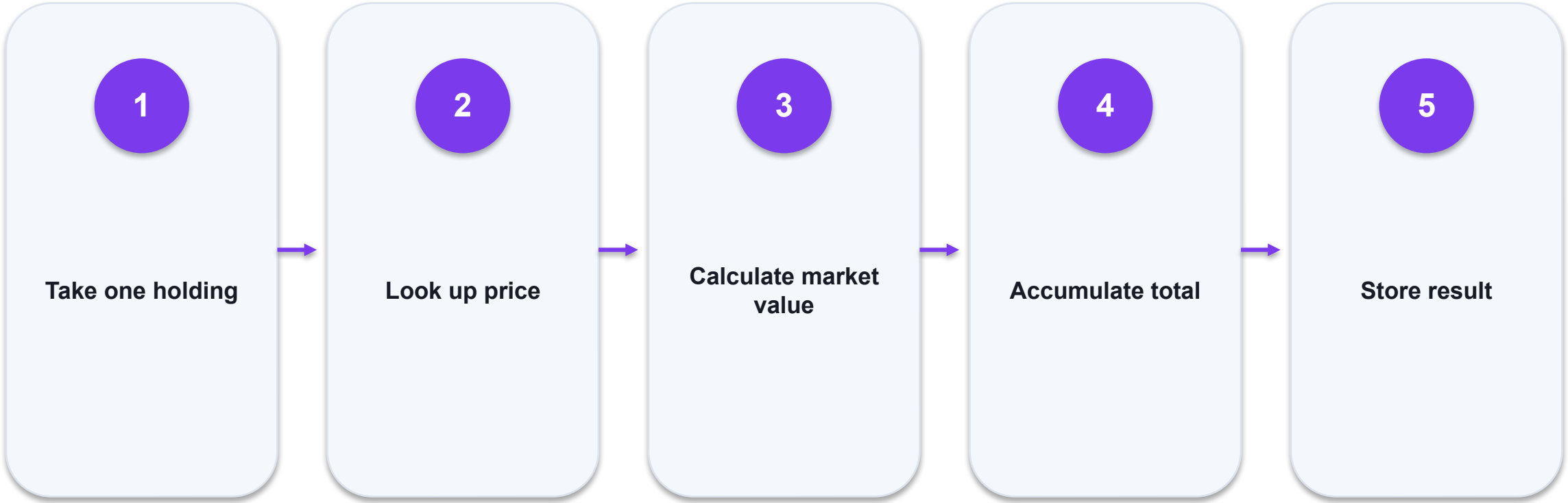
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Boundary conditions matter

Ambiguous	Explicit
PE < 15 is BUY and PE > 25 is SELL—but what happens at 15 and 25?	Define inclusive boundaries and test exactly below, at and above each threshold.

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For loops over a portfolio



The static slide contains the full sequence; trainers may reveal stages progressively.

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TOPIC 03 · COMPARISON

enumerate and zip improve clarity

enumerate	zip
Adds a stable row number while iterating.	Pairs aligned sequences such as tickers and quantities.

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TOPIC 03 · CONCEPT

break and continue are control signals

1

break

Stop after the required match or a risk limit breach.

2

continue

Skip a malformed record and log why it was excluded.

3

else on loop

Run only when the loop finishes without break.

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TOPIC 03 · DECISION FRAME

Comprehension or loop?

1

Simple transform

Comprehension is concise and expressive.

2

Multiple branches

A normal loop is easier to review.

3

Side effects

Use a loop; do not hide I/O in a comprehension.

4

Audit trail

Prefer clarity and intermediate variables.

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Thresholds need exact boundaries

```
1  if pe < 15: label = 'BUY'
2  elif pe <= 25: label = 'HOLD'
3  else: label = 'REVIEW'
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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A readable comprehension

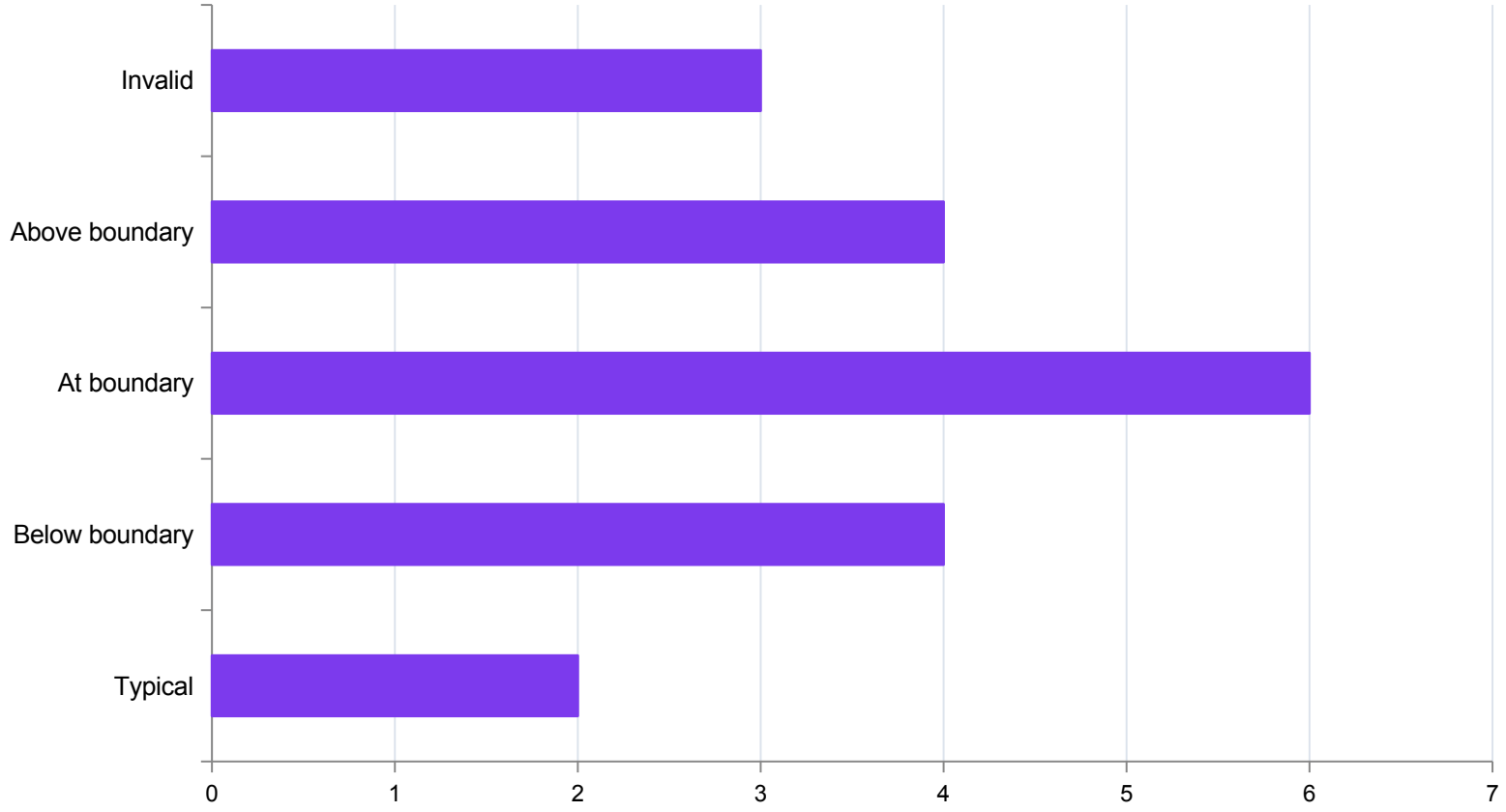
```
1  overweight = [h['ticker'] for h in holdings
2                if h['weight'] > 0.20]
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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Boundary tests concentrate around thresholds



WHAT THE DATA SHOWS

More tests belong at decision boundaries than in routine middle cases.

Recommended test cases by class

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WHAT YOU WILL APPLY

Hands-On Labs — Problem Solving with Control Structures

1

Lab 05

Encode PE and ROE Valuation Rules

2

Lab 06

Loop Through a Portfolio and Build Comprehensions

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Encode PE and ROE Valuation Rules



FINANCE SCENARIO

An equity screen must classify securities consistently at defined PE and ROE thresholds.

YOU WILL PRODUCE

A boundary-tested BUY/HOLD/REVIEW valuation classifier.

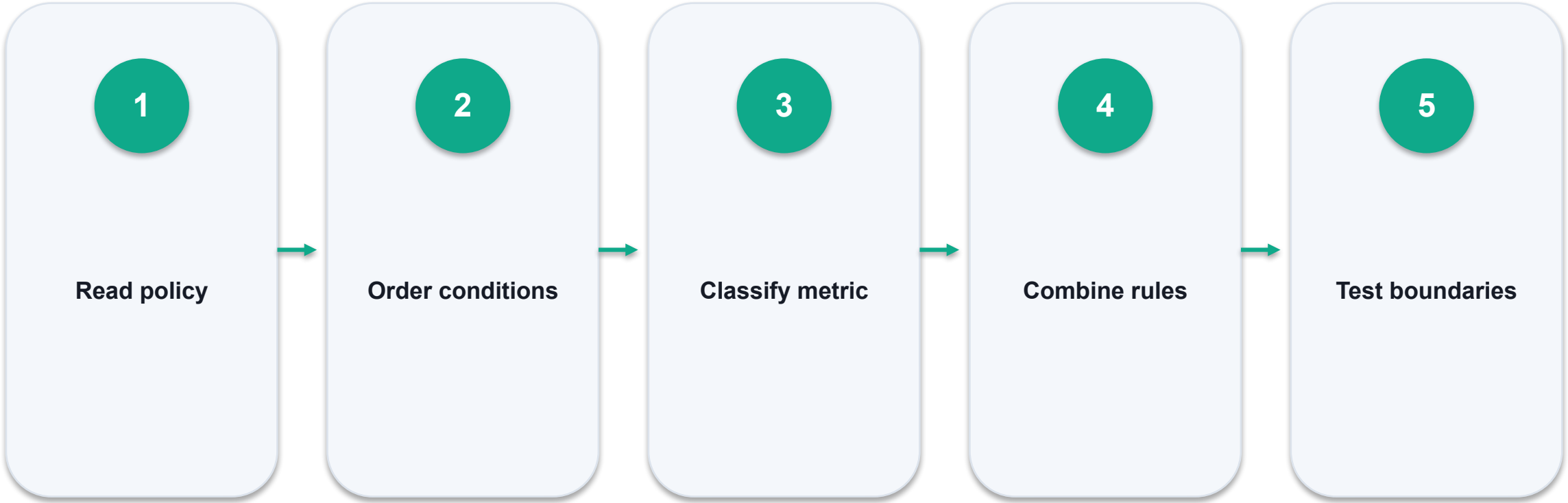
CONTROL **Apply PE and ROE boundaries explicitly, including exact thresholds and missing metrics.**

Python, unittest · Folder: labs/lab-05-*

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LAB 05 · CONCEPT MAP (DETAILED PROCEDURE IN THE LEARNER GUIDE)

Encode PE and ROE Valuation Rules — Analytical Workflow



The static slide contains the full sequence; trainers may reveal stages progressively.

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LAB 05 · VERIFY

Encode PE and ROE Valuation Rules — Evidence & Acceptance

ACCEPTANCE TEST

All exact-boundary tests pass and each recommendation includes the rule that produced it.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC 03 · HANDS-ON · LAB 06

Loop Through a Portfolio and Build Comprehensions

FINANCE SCENARIO

A portfolio analyst needs valuations for every valid position while malformed records are logged and excluded.

YOU WILL PRODUCE

A reconciled holdings table, exception log and concise list/dict comprehensions.

TOOLS & EVIDENCE

Python, csv

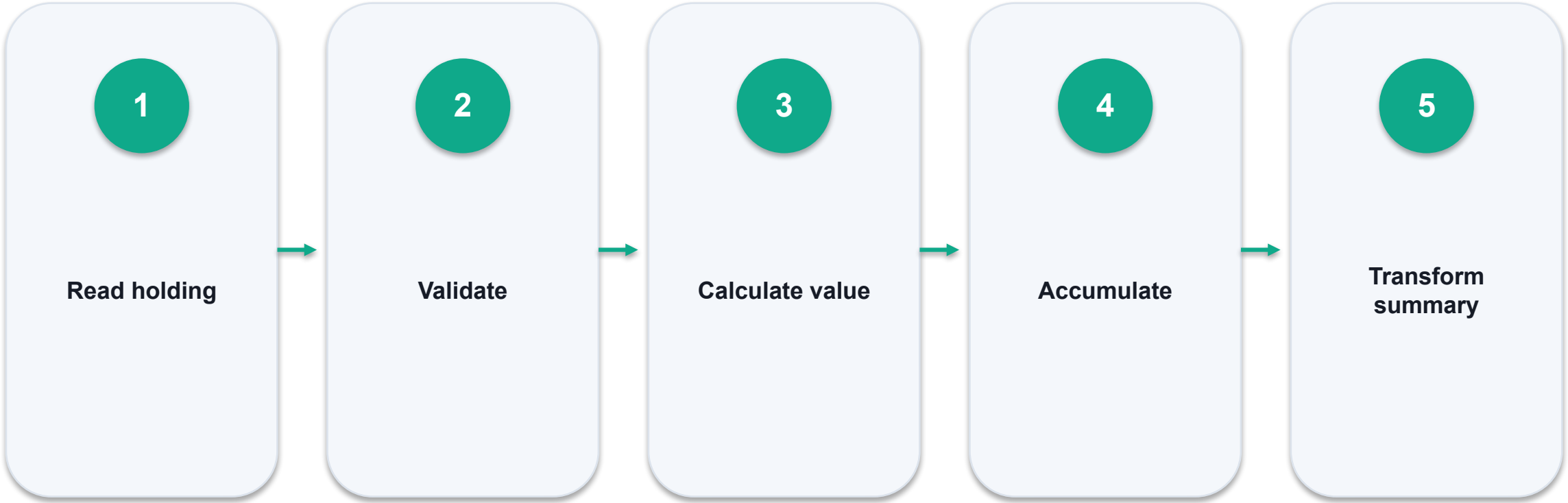
ALIGNMENT

LO3

Folder: labs/lab-06-*

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Loop Through a Portfolio and Build Comprehensions — Analytical Workflow



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LAB 06 · VERIFY

Loop Through a Portfolio and Build Comprehensions — Evidence & Acceptance

ACCEPTANCE TEST

The invalid row is logged once, valid positions reconcile to the portfolio total, and comprehension results match the loop results.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC RECAP

Recap — Problem Solving with Control Structures

1

Concept

Use if/elif/else when a finance rule selects one outcome from alternatives.

2

Method

Turn policy thresholds into explicit branches

3

Application

Lab 5, Lab 6

4

Evidence

Validate outputs and explain the finance meaning.

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02

START OF DAY 2

Day 2

Reusable code and Pandas data preparation

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DAY 2 · ORIENTATION

Day 2 Learning Route

1

Reconnect

Review the previous day’s evidence and open questions.

2

Concepts

Build the mental model before touching the code.

3

Application

Use finance scenarios and offline-safe data.

4

Evidence

Reconcile results and explain what they mean.

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04

TOPIC 04

Scripting with Functions and Lambda

Functions · parameters · return values · scope · lambda · map · filter

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LO4 · CONCEPT MAP

Key Concepts — Scripting with Functions and Lambda

1

Single purpose

A function should perform one understandable analytical responsibility.

2

Inputs and outputs

Parameters make assumptions visible; return values keep computation separate from display.

3

Reuse

Well-named functions reduce repeated formulas and inconsistent fixes.

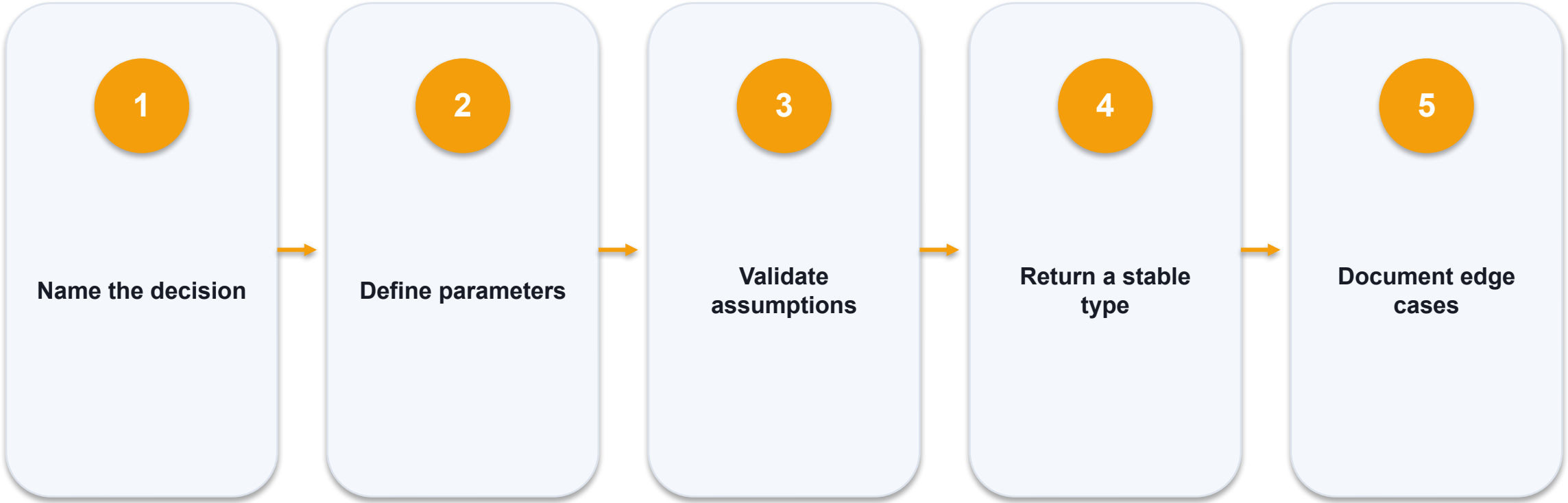
4

Lambda

Use small anonymous functions for short transformations, not complex business rules.

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Designing a finance function contract



The static slide contains the full sequence; trainers may reveal stages progressively.

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Parameters communicate intent

1

Positional

Compact when the order is obvious.

2

Keyword

Clearer for rates, periods and optional settings.

3

Default

Useful only when a defensible standard exists.

4

Variable arguments

Appropriate for flexible aggregation, used carefully.

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Return calculations; print presentation

Return	Print
The result can be tested, reused, stored or charted.	Creates display output but is difficult to compose into another calculation.

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Lambda, map and filter



The static slide contains the full sequence; trainers may reveal stages progressively.

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TOPIC 04 · COMPARISON

Readable functions beat clever one-liners

Named function	Lambda
Best for multi-step logic, validation, documentation and tests.	Best for one short expression close to where it is used.

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Unit tests protect recurring reports

1

Typical

A representative input produces the known value.

2

Boundary

Zero, exactly-at-threshold and extreme values behave correctly.

3

Invalid

Bad inputs fail safely with a useful message.

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Return values keep logic reusable

```
1  def roe(net_income, avg_equity):
2      return None if avg_equity == 0 else net_income / avg_equity
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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Lambda stays small

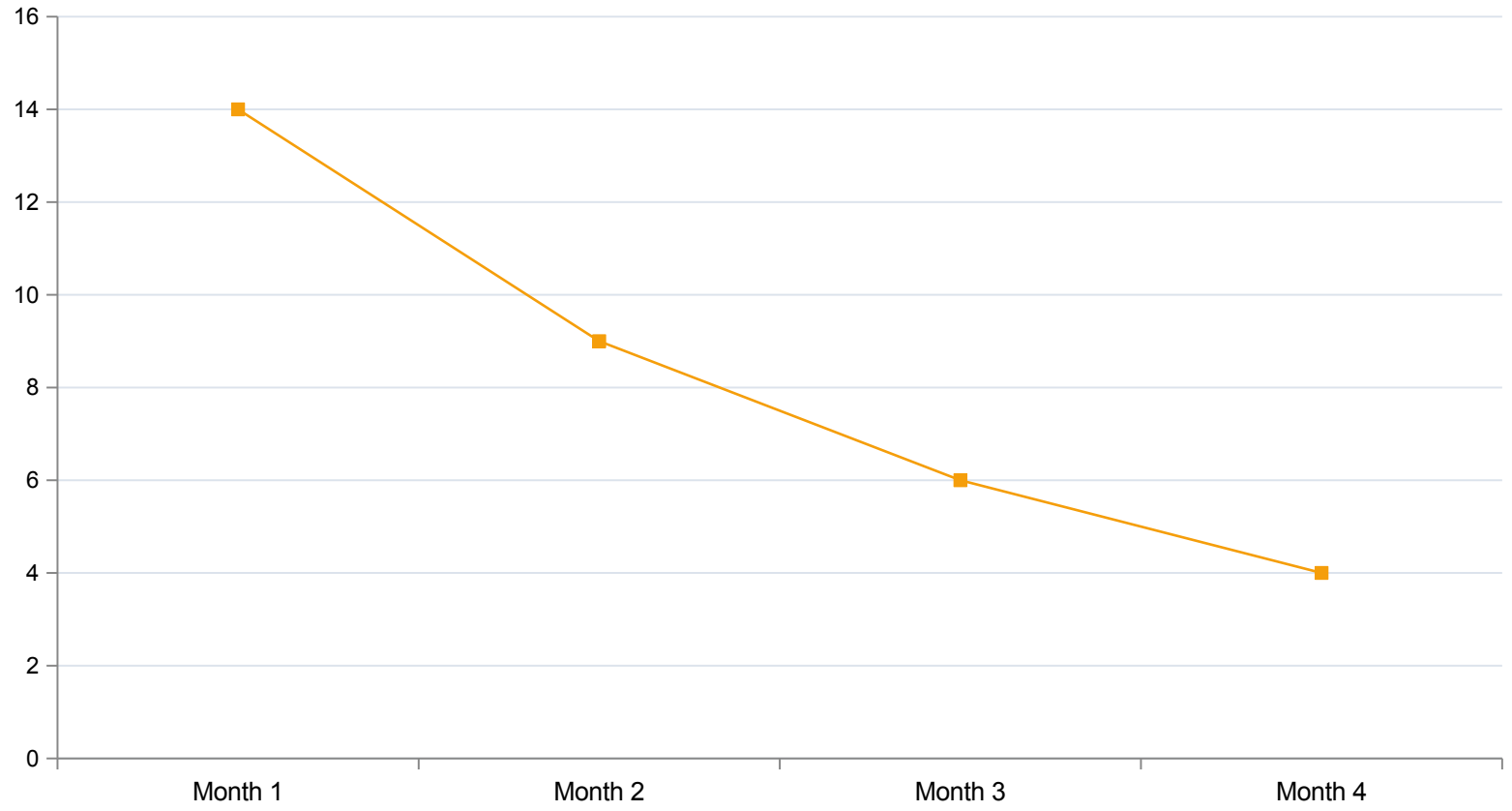
```
high_dti = filter(lambda row: row['dti'] > 0.45, applicants)
manual_review = list(high_dti)
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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Reusable functions reduce duplicate formulas



WHAT THE DATA SHOWS

Centralising formulas reduces inconsistent fixes and review effort.

Duplicate calculation blocks found

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WHAT YOU WILL APPLY

Hands-On Labs — Scripting with Functions and Lambda

1

Lab 07

Create Reusable Financial Ratio Functions

2

Lab 08

Screen Customers with Lambda, Map and Filter

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Create Reusable Financial Ratio Functions

FINANCE SCENARIO

A bank reporting team calculates margin, ROE and ROA in several reports and needs one tested implementation.

YOU WILL PRODUCE

A ratios module returning named results with validation and tests.

TOOLS & EVIDENCE

Python, dataclasses, unittest

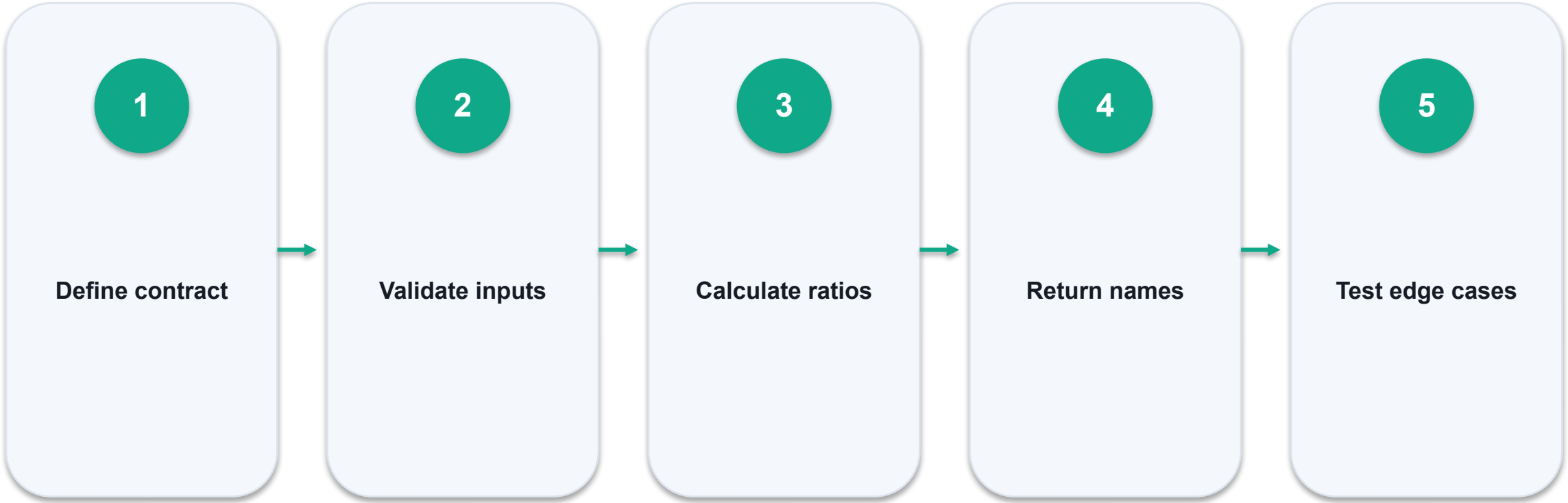
ALIGNMENT

LO4

Folder: labs/lab-07-*

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Create Reusable Financial Ratio Functions — Analytical Workflow



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LAB 07 · VERIFY

Create Reusable Financial Ratio Functions — Evidence & Acceptance

ACCEPTANCE TEST

The sample produces margin about 55.95%, ROE about 18.09% and ROA about 1.53%; tests pass.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC 04 · HANDS-ON · LAB 08

Screen Customers with Lambda, Map and Filter



FINANCE SCENARIO

A retail bank wants to transform customer income into affordability ratios and filter applications for manual review.

YOU WILL PRODUCE

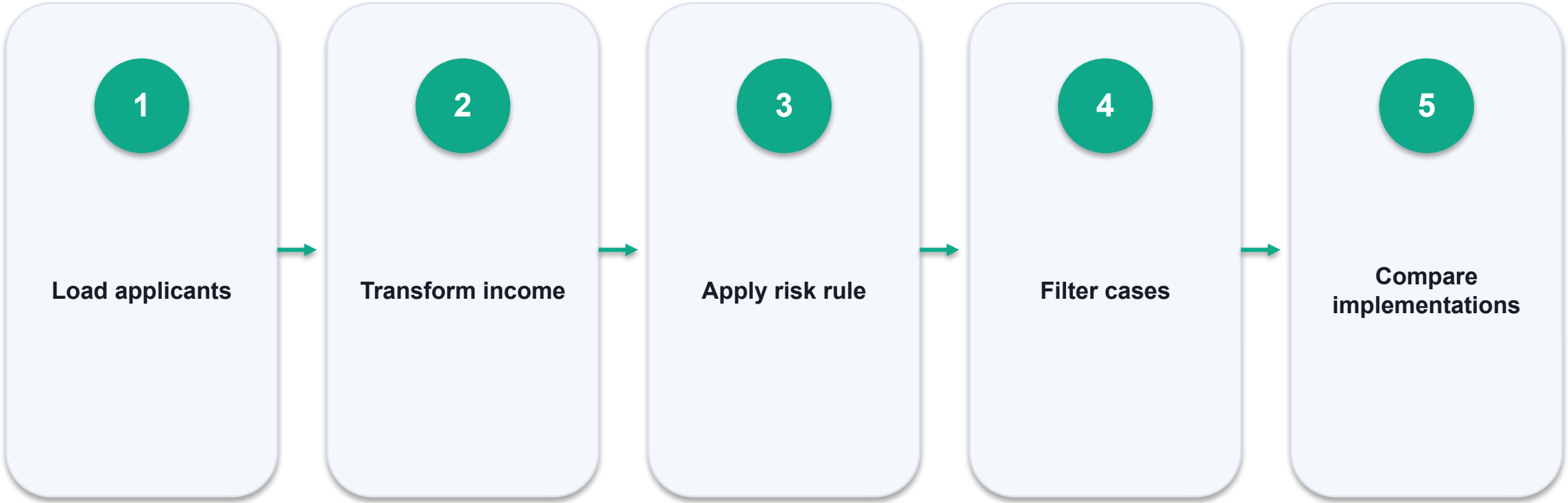
A readable transformation pipeline plus an equivalent named-function implementation.

CONTROL **Keep segmentation rules reviewable and record every applicant routed to manual review.**

Python, csv, unittest · Folder: labs/lab-08-*

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Screen Customers with Lambda, Map and Filter — Analytical Workflow



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LAB 08 · VERIFY

Screen Customers with Lambda, Map and Filter — Evidence & Acceptance

ACCEPTANCE TEST

Lambda and named-function pipelines select the same applicant IDs, and counts reconcile to the input.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC RECAP

Recap — Scripting with Functions and Lambda

1

Concept

A function should perform one understandable analytical responsibility.

2

Method

Designing a finance function contract

3

Application

Lab 7, Lab 8

4

Evidence

Validate outputs and explain the finance meaning.

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05

TOPIC 05

Import and Process Finance Data

Pandas · Series · DataFrame · import · filter · slice · clean

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LO5 · CONCEPT MAP

Key Concepts — Import and Process Finance Data

1

Series

A labelled one-dimensional array, such as closing prices indexed by date.

2

DataFrame

A labelled table of rows and columns suitable for finance records and time series.

3

Selection

Use labels and conditions deliberately with loc, iloc and boolean masks.

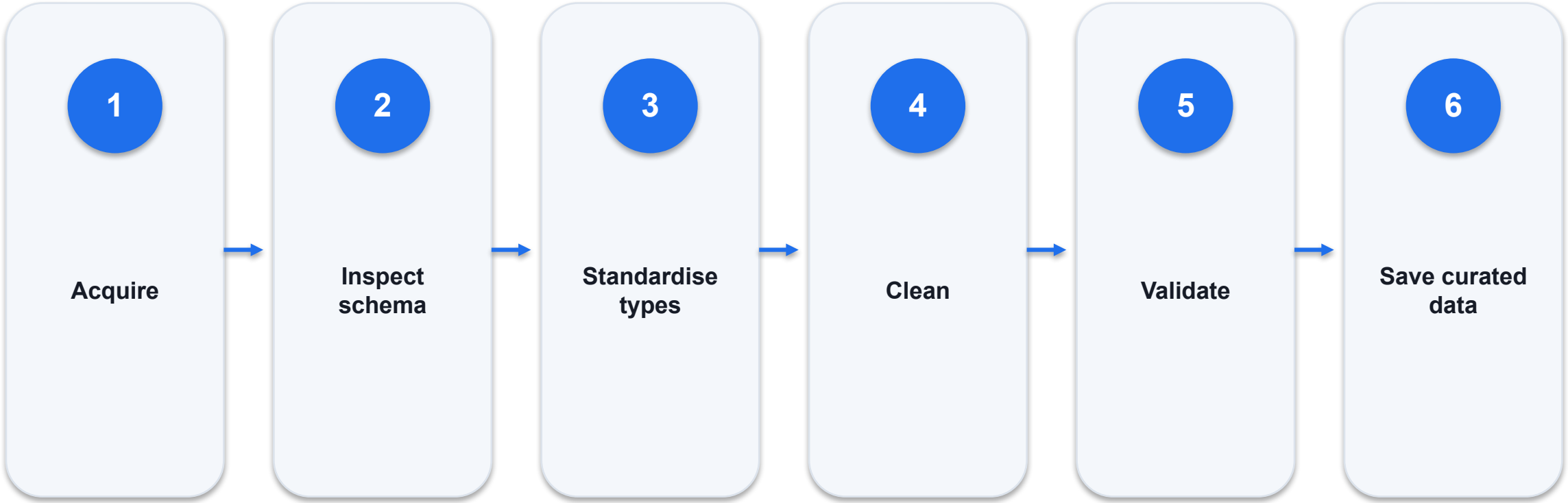
4

Data quality

Missing, duplicate, invalid and inconsistent values must be measured before analysis.

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The finance data pipeline



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TOPIC 05 · COMPARISON

Series and DataFrame mental model

Series	DataFrame
One named measure plus an index.	Aligned Series sharing a row index.

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TOPIC 05 · CONCEPT

Import defensively

1

CSV

Specify dates, dtypes and missing-value tokens where practical.

2

Excel

Select the intended sheet and confirm header rows.

3

API

Handle empty responses, rate limits and changing fields.

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TOPIC 05 · COMPARISON

loc and iloc answer different questions

loc	iloc
Select by label or boolean condition; endpoints are label-inclusive.	Select by integer position; upper bound is excluded like normal slicing.

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TOPIC 05 · DECISION FRAME

Missing data is a business question

1

Drop

Valid when incomplete rows cannot support the decision.

2

Fixed fill

Use only when zero or another constant has business meaning.

3

Forward fill

Carries the last observed value; risky across long gaps.

4

Impute/model

Requires assumptions that must be documented and tested.

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Data-quality controls



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Inspect before transforming

```
1  df = pd.read_csv('prices.csv', parse_dates=['Date'])
2  print(df.shape, df.dtypes, df.isna().sum())
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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TOPIC 05 · WORKED EXAMPLE

Label selection states the business rule

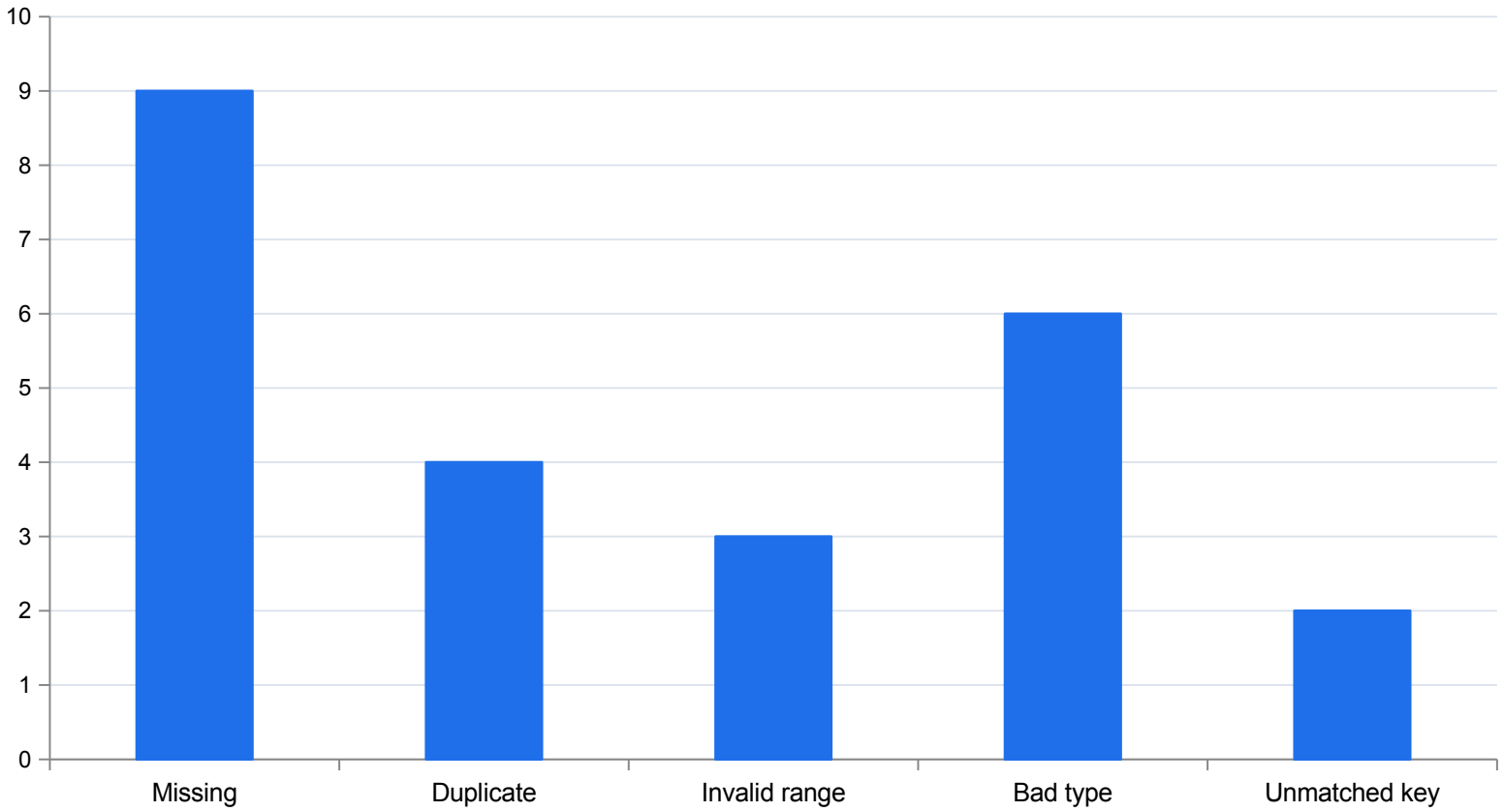
```
banks = portfolio.loc[
    (portfolio['Sector'] == 'Bank') &
(portfolio['Value'] > 100_000)
]
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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A data audit makes defects visible



WHAT THE DATA SHOWS

Profiling defects before cleaning creates a defensible reconciliation.

Exceptions detected

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WHAT YOU WILL APPLY

Hands-On Labs — Import and Process Finance Data

1

Lab 09

Import and Clean a Finance DataFrame

2

Lab 10

Filter and Slice Finance Data Safely

3

Lab 11

Run a Finance Data-Quality Audit

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TOPIC 05 · HANDS-ON · LAB 09

Import and Clean a Finance DataFrame

FINANCE SCENARIO

A treasury extract contains duplicated dates, missing prices and inconsistent date/value types.

YOU WILL PRODUCE

A curated price DataFrame and a written data-quality reconciliation.

TOOLS & EVIDENCE

Pandas, CSV

ALIGNMENT

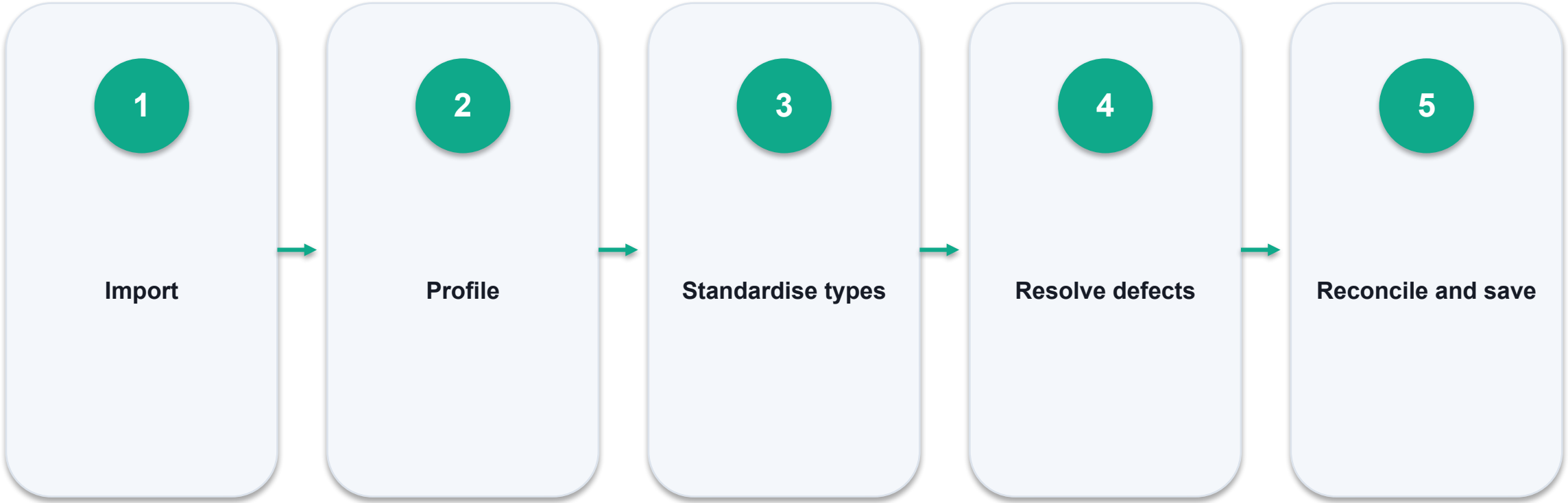
LO5

Folder: labs/lab-09-*

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LAB 09 · CONCEPT MAP (DETAILED PROCEDURE IN THE LEARNER GUIDE)

Import and Clean a Finance DataFrame — Analytical Workflow



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LAB 09 · VERIFY

Import and Clean a Finance DataFrame — Evidence & Acceptance

ACCEPTANCE TEST

The curated file has unique sorted dates, numeric prices, no missing Close values, and a reconciliation from raw to clean rows.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC 05 · HANDS-ON · LAB 10

Filter and Slice Finance Data Safely

FINANCE SCENARIO

A portfolio manager asks for bank-sector positions above a materiality threshold and a date-windowed price series.

YOU WILL PRODUCE

A filtered positions file and labelled versus positional slice comparison.

TOOLS & EVIDENCE

Pandas, CSV

ALIGNMENT

LO5

Folder: labs/lab-10-*

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Filter and Slice Finance Data Safely — Analytical Workflow



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LAB 10 · VERIFY

Filter and Slice Finance Data Safely — Evidence & Acceptance

ACCEPTANCE TEST

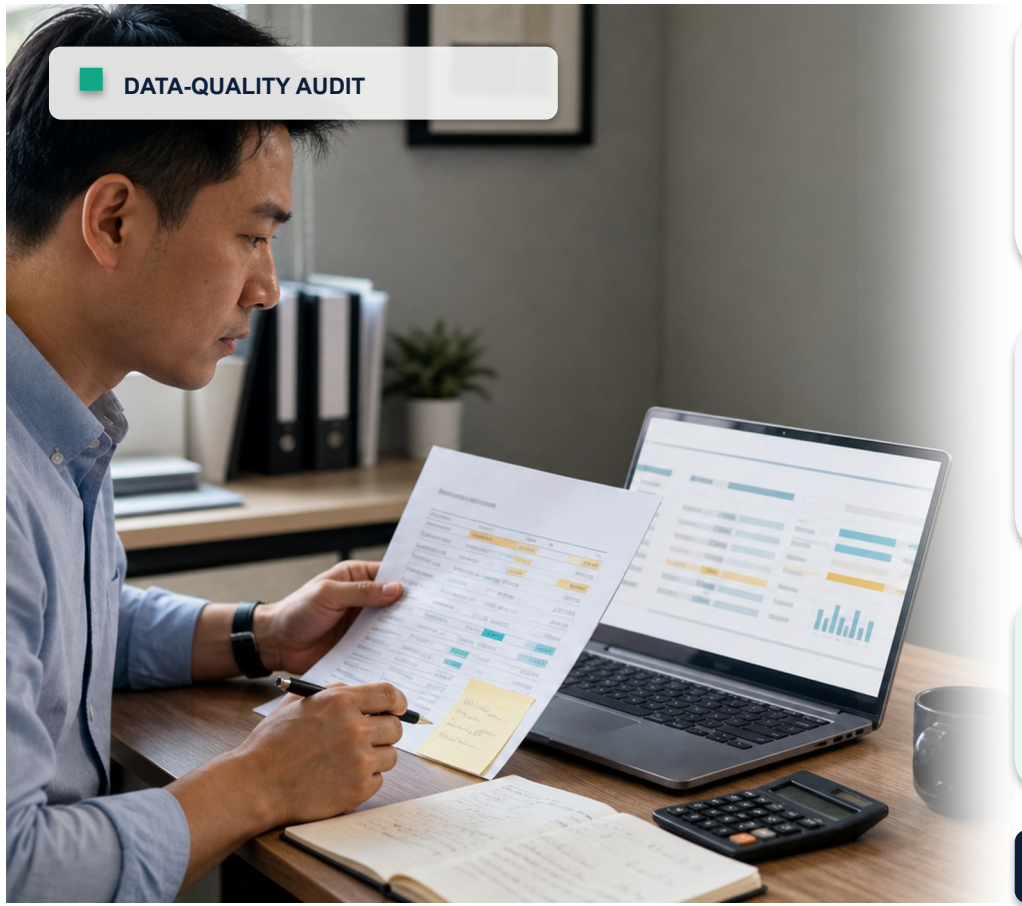
The filter returns only qualifying bank positions; loc and iloc row counts match their documented boundary rules.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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Run a Finance Data-Quality Audit



FINANCE SCENARIO

A credit portfolio extract must pass completeness, uniqueness, range and reconciliation controls before modelling.

YOU WILL PRODUCE

A reusable audit function and exception report for loan records.

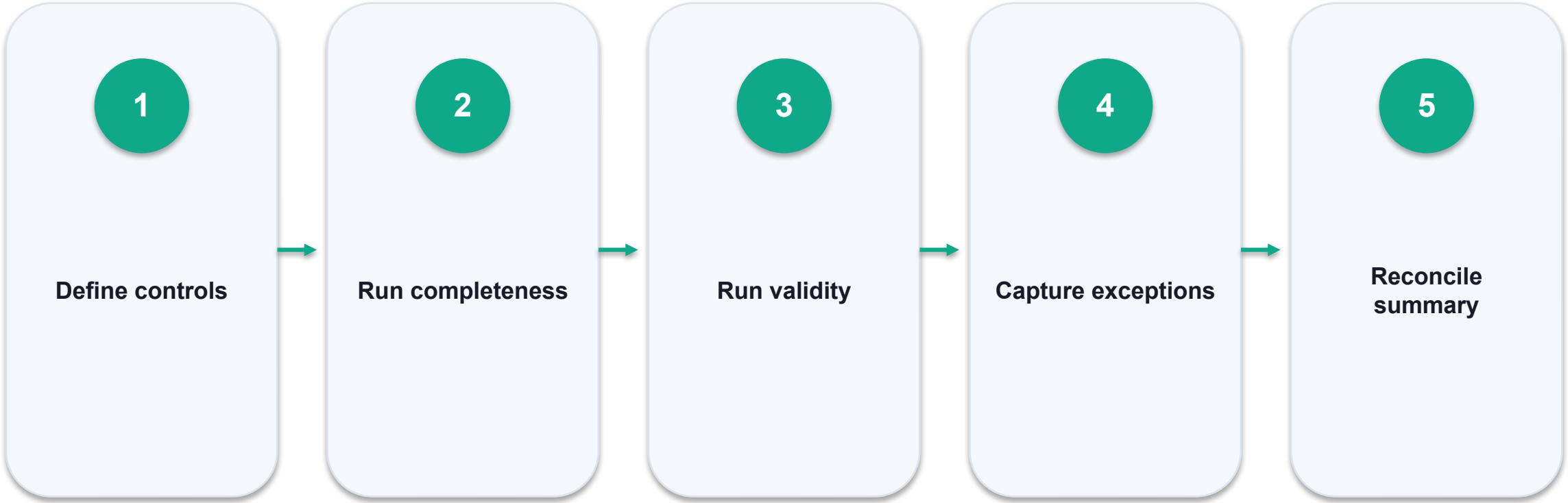
CONTROL **Fail the run when row counts, unique IDs, ranges or reconciled totals breach agreed controls.**

Pandas, CSV, JSON · Folder: labs/lab-11-*

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LAB 11 · CONCEPT MAP (DETAILED PROCEDURE IN THE LEARNER GUIDE)

Run a Finance Data-Quality Audit — Analytical Workflow



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LAB 11 · VERIFY

Run a Finance Data-Quality Audit — Evidence & Acceptance

ACCEPTANCE TEST

The audit identifies every planted defect, produces machine-readable counts and leaves the raw file unchanged.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC RECAP

Recap — Import and Process Finance Data

1

Concept

A labelled one-dimensional array, such as closing prices indexed by date.

2

Method

The finance data pipeline

3

Application

Lab 9, Lab 10, Lab 11

4

Evidence

Validate outputs and explain the finance meaning.

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START OF DAY 3

Day 3

Aggregation, visualisation and financial analysis

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DAY 3 · ORIENTATION

Day 3 Learning Route

1

Reconnect

Review the previous day’s evidence and open questions.

2

Concepts

Build the mental model before touching the code.

3

Application

Use finance scenarios and offline-safe data.

4

Evidence

Reconcile results and explain what they mean.

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06

TOPIC 06

Aggregate and Visualize Finance Data

concat · merge · groupby · pivot table · returns · charts

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LO6 · CONCEPT MAP

Key Concepts — Aggregate and Visualize Finance Data

1

Combine

concat stacks compatible data; merge joins tables using explicit keys.

2

Aggregate

groupby splits records, applies calculations and combines results.

3

Reshape

Pivot tables summarise a measure across two or more dimensions.

4

Visualise

Choose a chart that matches the analytical question and label units clearly.

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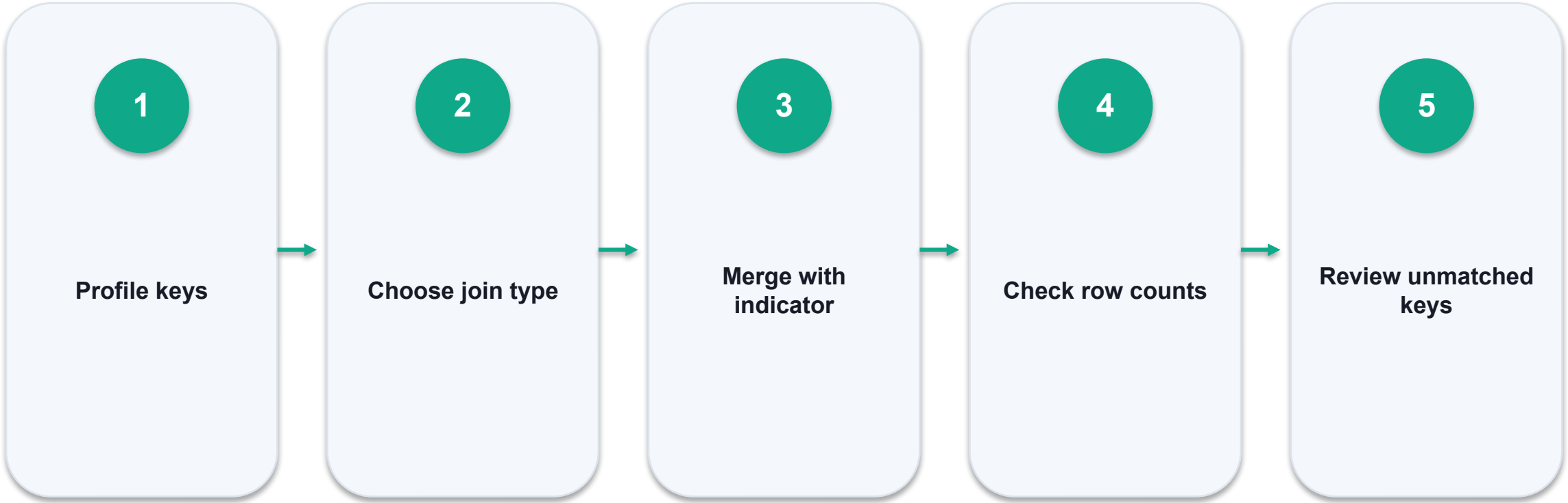
TOPIC 06 · COMPARISON

concat and merge solve different problems

concat	merge
Append periods or place aligned series side by side.	Join facts and reference data by a validated key.

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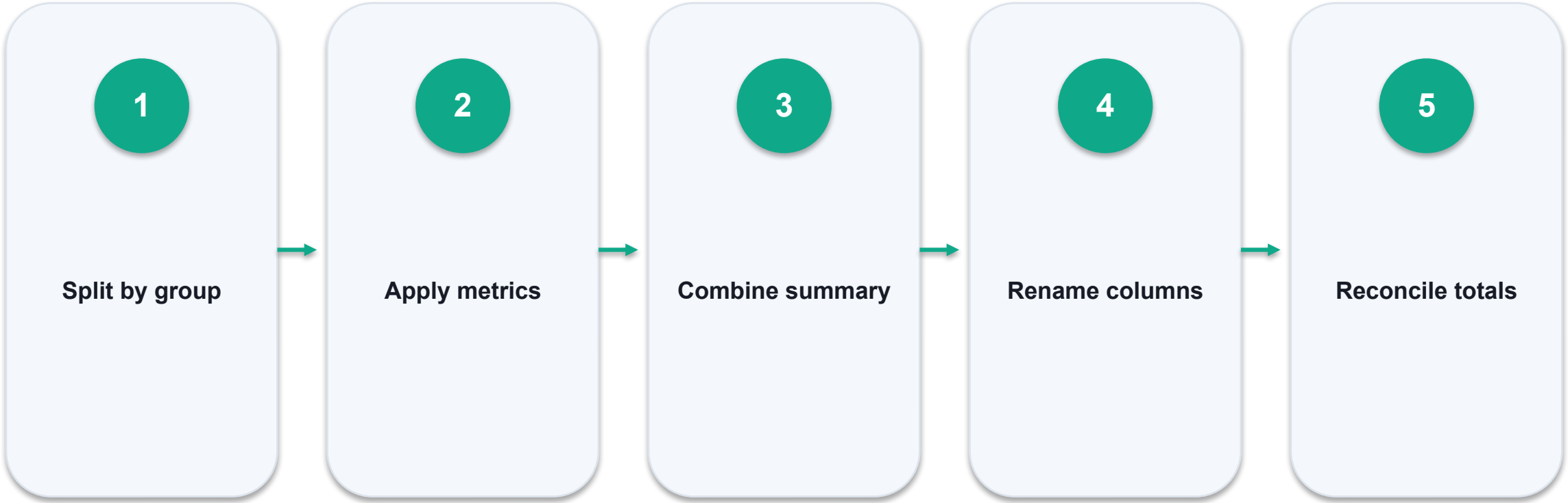
Validate every join



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The split-apply-combine model



The static slide contains the full sequence; trainers may reveal stages progressively.

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Pivot tables answer cross-dimensional questions

1 **Index**

Rows, such as sector.

2 **Columns**

Categories, such as risk band.

3 **Values**

Measure, such as exposure.

4 **aggfunc**

How duplicates are summarised.

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Choose the chart for the question

1

Line

Trend or comparison through time.

2

Bar

Compare categories.

3

Scatter

Relationship between two measures.

4

Histogram

Distribution and unusual observations.

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TOPIC 06 · CONCEPT

A chart is an analytical claim

1

Context

State date range, currency, population and data source.

2

Scale

Use honest axes and visible units.

3

Takeaway

Explain what changed, why it matters and what to check next.

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Validate the merge relationship

```
merged = apps.merge(sectors, on='Sector_Code',
                    how='left', validate='many_to_one',
indicator=True)
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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Returns turn price levels into change

```
1  prices['Daily_Return'] = prices['Close'].pct_change()

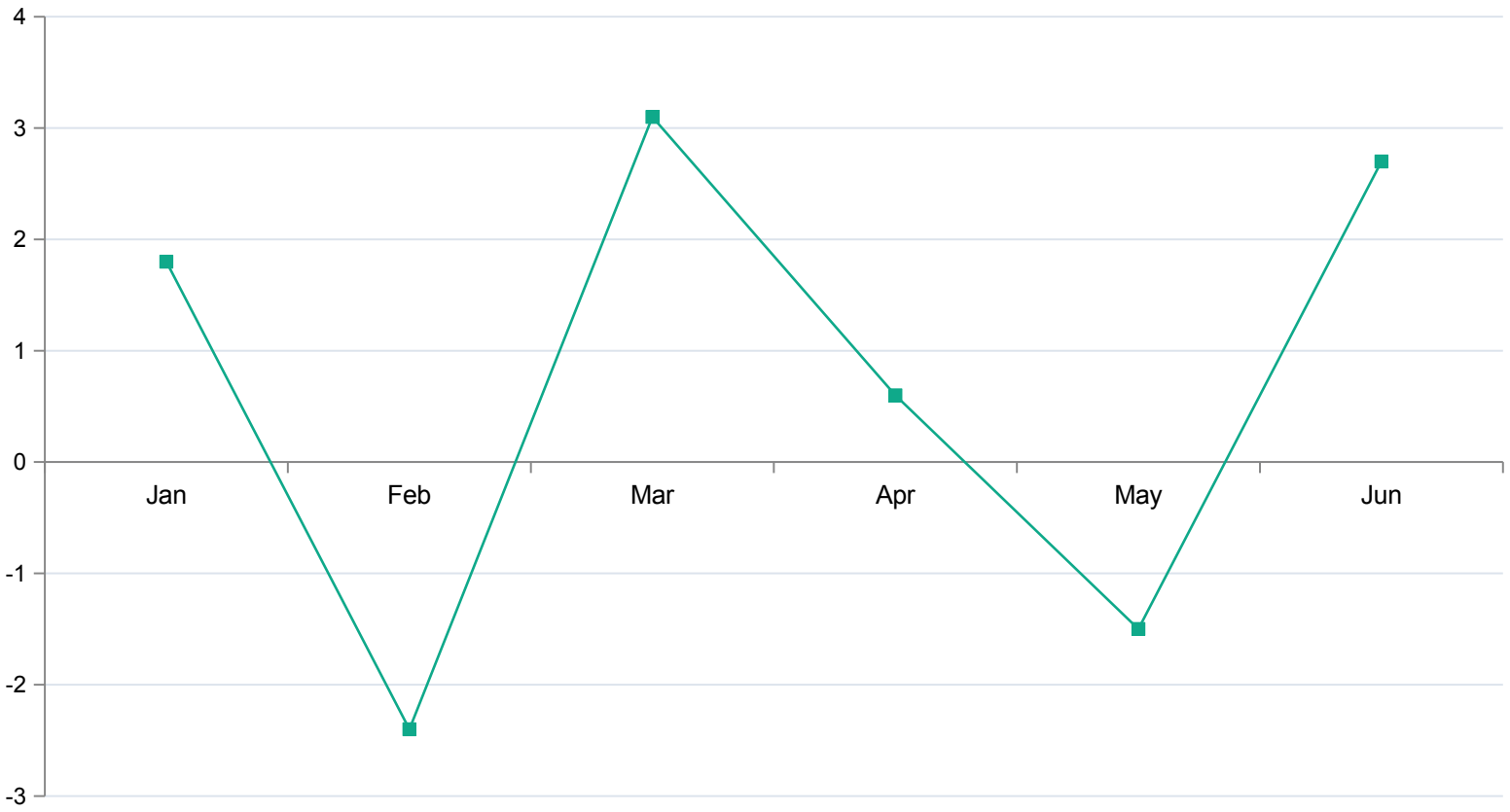
2  vol = prices['Daily_Return'].std() * (252 ** 0.5)
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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Returns reveal risk hidden by price levels



WHAT THE DATA SHOWS

A smooth price path can still contain volatile returns and tail observations.

Illustrative monthly return (%)

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WHAT YOU WILL APPLY

Hands-On Labs — Aggregate and Visualize Finance Data

1

Lab 12

Join, Group and Pivot a Loan Portfolio

2

Lab 13

Visualize Daily Returns and Distribution

3

Lab 14

Build a Moving-Average Technical Analysis

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Join, Group and Pivot a Loan Portfolio



FINANCE SCENARIO

A risk team needs exposure and approval metrics by sector, region and risk band from separate application and reference tables.

YOU WILL PRODUCE

A validated merge, groupby summary and pivot table exported to CSV.

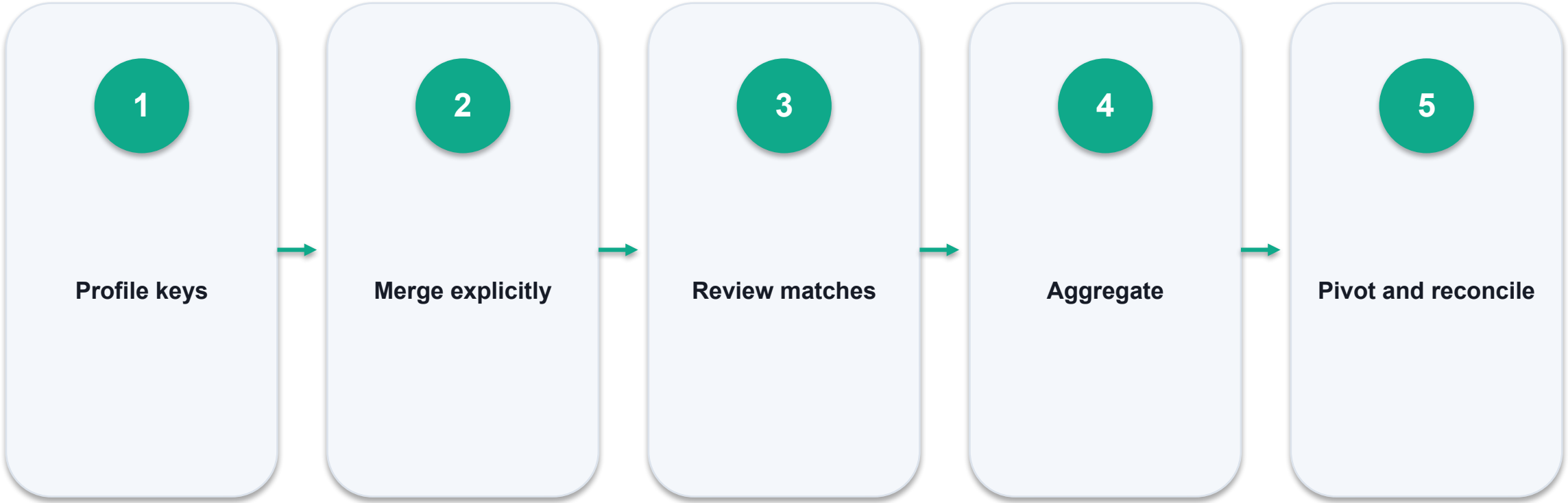
CONTROL **Validate the merge relationship, unmatched keys and reconciled totals before trusting the pivot.**

Pandas, CSV · Folder: labs/lab-12-*

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LAB 12 · CONCEPT MAP (DETAILED PROCEDURE IN THE LEARNER GUIDE)

Join, Group and Pivot a Loan Portfolio — Analytical Workflow



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LAB 12 · VERIFY

Join, Group and Pivot a Loan Portfolio — Evidence & Acceptance

ACCEPTANCE TEST

No unexpected many-to-many join occurs, unmatched keys are reported, and summary exposure totals equal the source.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC 06 · HANDS-ON · LAB 13

Visualize Daily Returns and Distribution

FINANCE SCENARIO

An investment committee wants to understand daily price changes and the distribution of returns for DBS.

YOU WILL PRODUCE

A return series, line chart and histogram with labelled analytical takeaways.

TOOLS & EVIDENCE

Pandas, Matplotlib, CSV

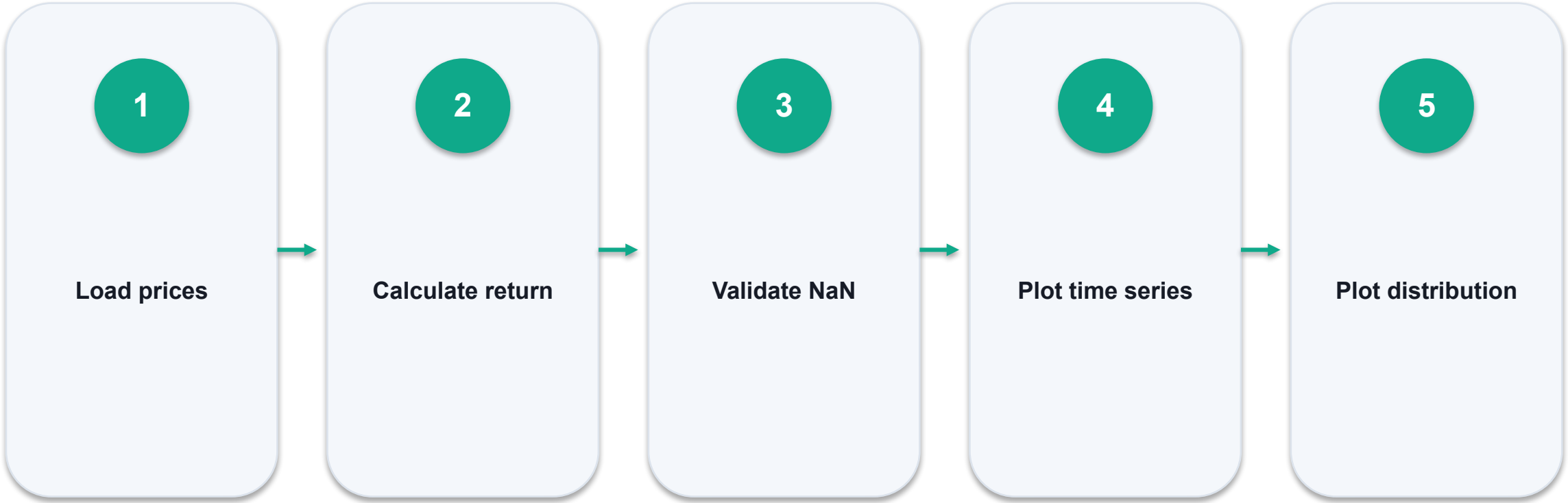
ALIGNMENT

LO6 / A4 / A5

Folder: labs/lab-13-*

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Visualize Daily Returns and Distribution — Analytical Workflow



The static slide contains the full sequence; trainers may reveal stages progressively.

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LAB 13 · VERIFY

Visualize Daily Returns and Distribution — Evidence & Acceptance

ACCEPTANCE TEST

returns.csv has one fewer non-null row than prices; charts render; summary statistics match Pandas calculations.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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Build a Moving-Average Technical Analysis



FINANCE SCENARIO

A quantitative analyst needs an OHLC view and 10/50-day moving averages with objectively detected crossover dates.

YOU WILL PRODUCE

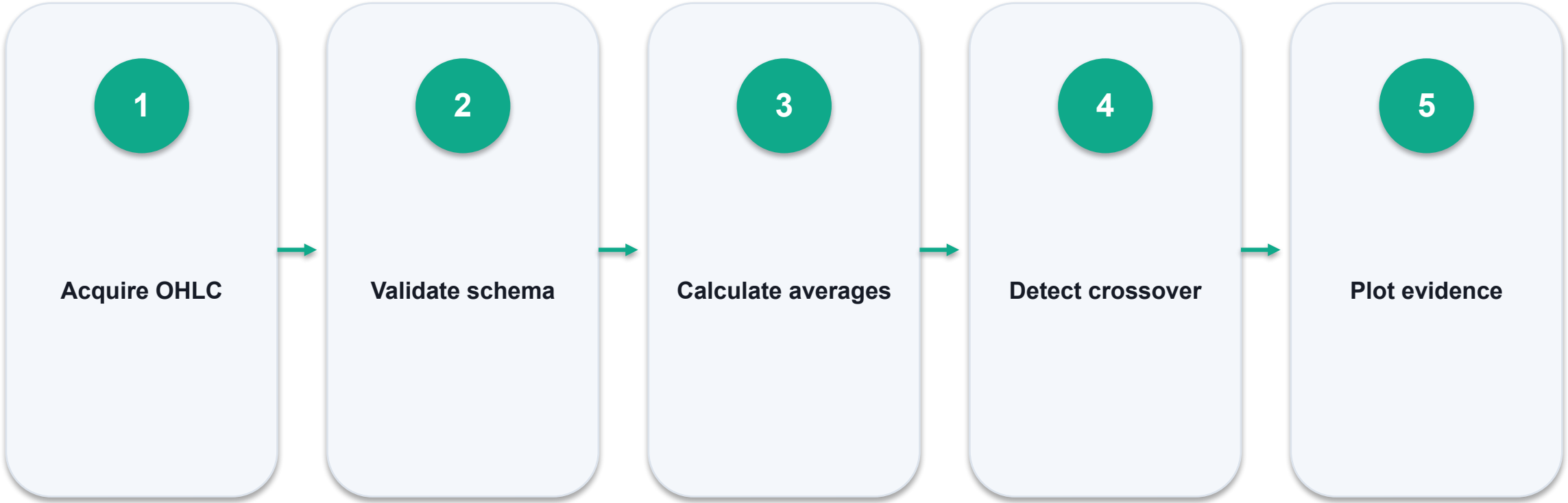
An offline-safe technical-analysis report with OHLC, moving averages and crossover evidence.

CONTROL **Use one price basis, verify both moving-average windows and label only detected crossovers.**

Pandas, Matplotlib, optional yfinance · Folder: labs/lab-14-*

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Build a Moving-Average Technical Analysis — Analytical Workflow



The static slide contains the full sequence; trainers may reveal stages progressively.

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LAB 14 · VERIFY

Build a Moving-Average Technical Analysis — Evidence & Acceptance

ACCEPTANCE TEST

The output contains OHLC rows, MA10/MA50 values, at least one crossover date, and a labelled chart; the offline file always works.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC RECAP

Recap — Aggregate and Visualize Finance Data

1

Concept

concat stacks compatible data; merge joins tables using explicit keys.

2

Method

concat and merge solve different problems

3

Application

Lab 12, Lab 13, Lab 14

4

Evidence

Validate outputs and explain the finance meaning.

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TOPIC 07

Analyze Finance Data

apply · pipe · statistics · correlation · rolling analysis · change tracking

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LO7 · CONCEPT MAP

Key Concepts — Analyze Finance Data

1

apply

Run a function across rows or columns when vectorised operations are not practical.

2

pipe

Express a readable sequence of named DataFrame transformations.

3

Statistics

Describe centre, spread, relationships and unusual observations before drawing conclusions.

4

Change over time

Returns, rolling windows and drawdown expose behaviour that levels alone can hide.

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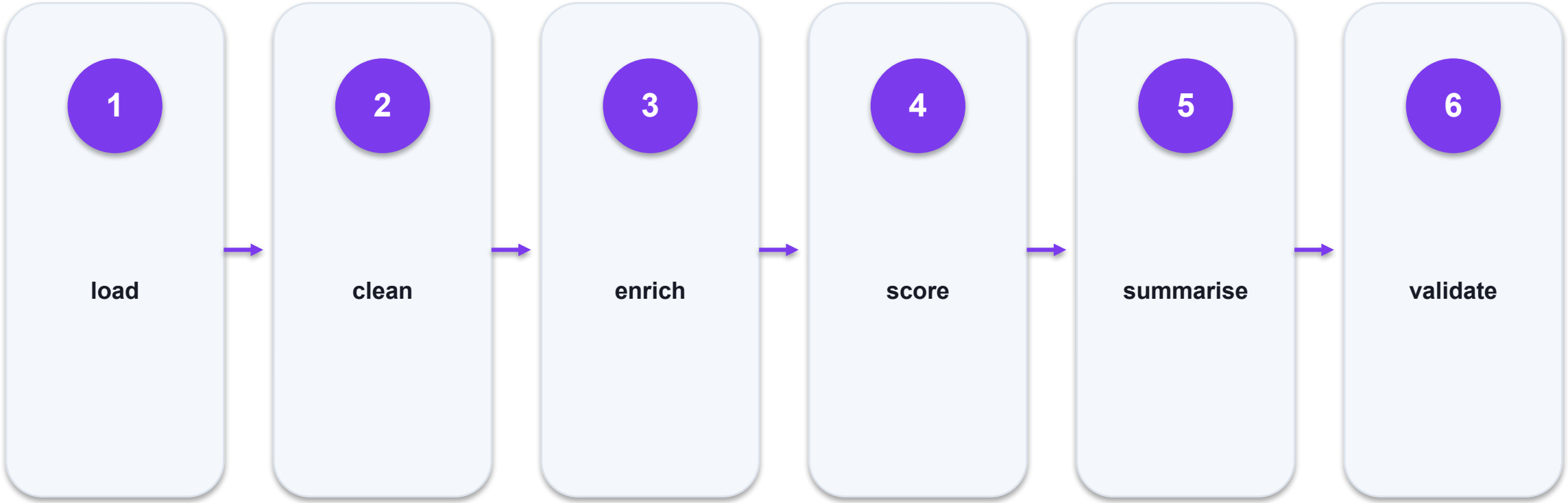
TOPIC 07 · COMPARISON

apply is flexible, but not automatically fast

Vectorised	apply
Preferred for arithmetic and conditions supported directly by Pandas/NumPy.	Useful for custom row logic; benchmark on large data.

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Build a readable pipe chain



The static slide contains the full sequence; trainers may reveal stages progressively.

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Descriptive statistics form a profile

1

Mean/median

Typical level and sensitivity to skew.

2

Std/range

Dispersion and volatility.

3

Quantiles

Distribution and tail thresholds.

4

Count/missing

Coverage and reliability.

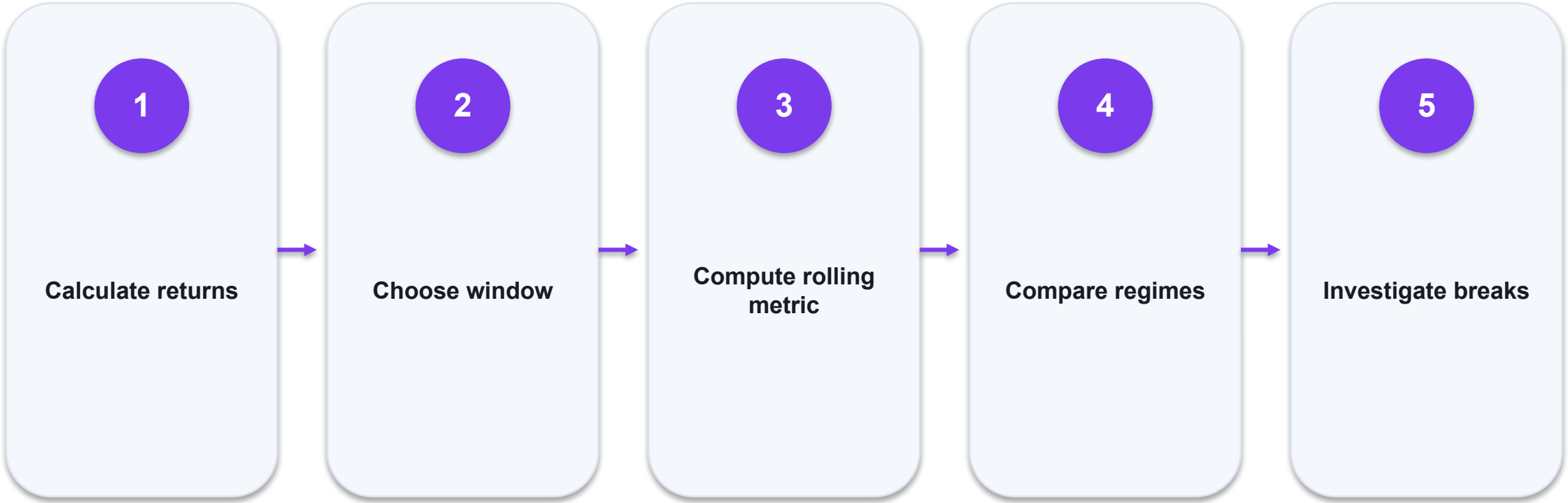
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Covariance and correlation

Covariance	Correlation
Direction and joint movement in scale-dependent units.	Normalised association from -1 to +1; not causation.

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Rolling windows reveal regime change



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Evidence before recommendation



1 Reconcile

Totals and row counts agree with source data.

2 Stress

Results hold across boundaries and plausible scenarios.

3 Explain

A reviewer can trace the output back to data and assumptions.

DECISION / CONTROL

A recommendation is usable only after source totals reconcile, assumptions are stressed and material limitations are explained.

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Pipe reads like the analytical story

```
result = (raw.pipe(clean_data)
             .pipe(add_features)
             .pipe(score_risk)
             .pipe(summarise))
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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Drawdown uses the running peak

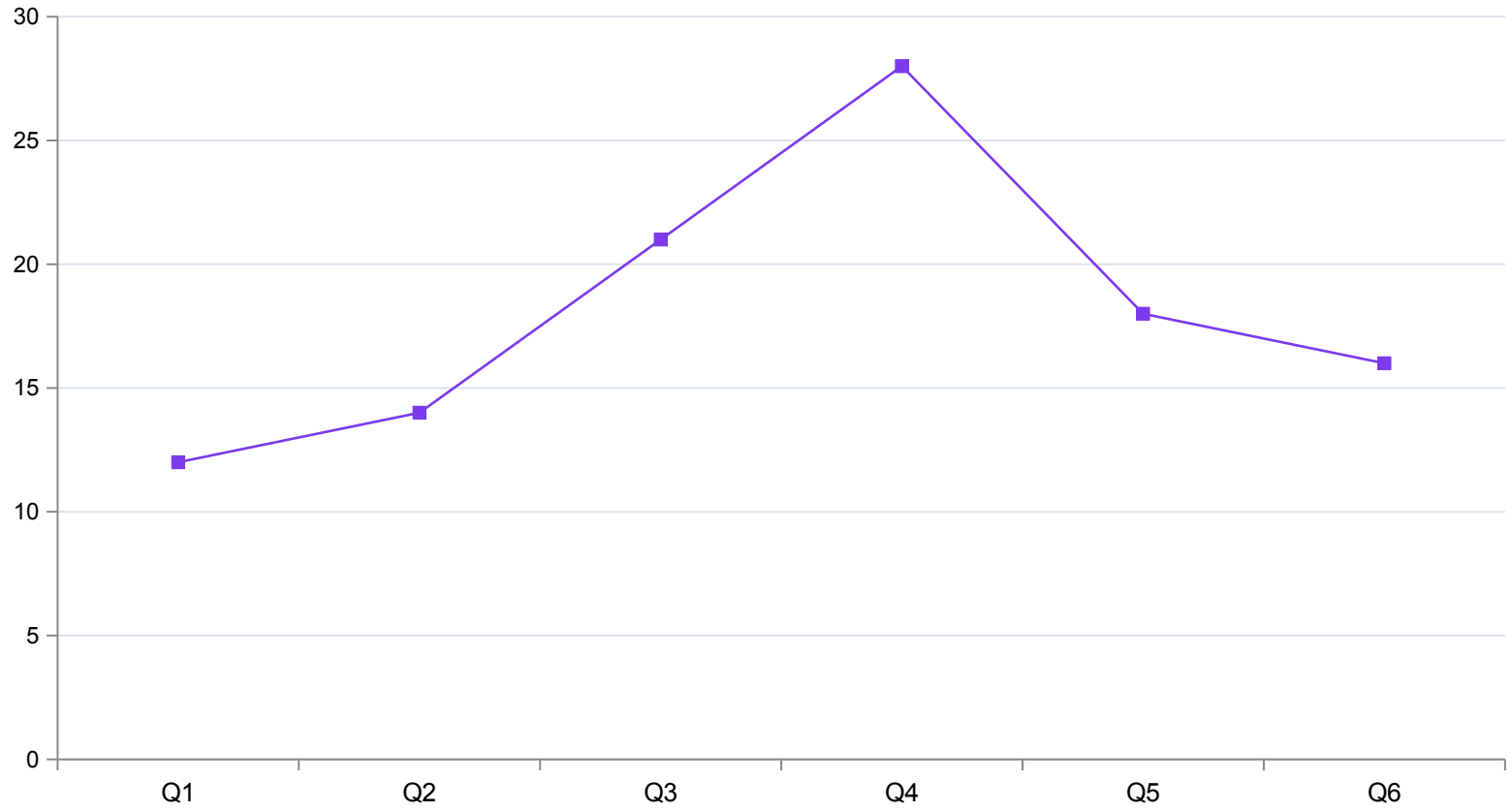
```
1  wealth = (1 + returns).cumprod()
2  peak = wealth.cummax()
3  drawdown = wealth / peak - 1
```

HOW TO READ IT

1. Inputs and assumptions are visible.
2. The calculation is separated from presentation.
3. Boundary or missing-data behaviour is explicit.
4. A reviewer can test the result independently.

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Rolling volatility changes across regimes



WHAT THE DATA SHOWS

Risk is time-varying; a single full-period volatility figure can hide regime changes.

Annualised volatility (%)

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WHAT YOU WILL APPLY

Hands-On Labs — Analyze Finance Data

1

Lab 15

Build a Clean-to-Risk Pipeline with pipe

2

Lab 16

Analyse Statistics, Correlation and Rolling Risk

3

Lab 17

Capstone: Produce an Executive Finance Analytics Pack

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Build a Clean-to-Risk Pipeline with pipe



FINANCE SCENARIO

A credit-risk team wants the DataFrame transformations to read like an auditable sequence rather than one large notebook cell.

YOU WILL PRODUCE

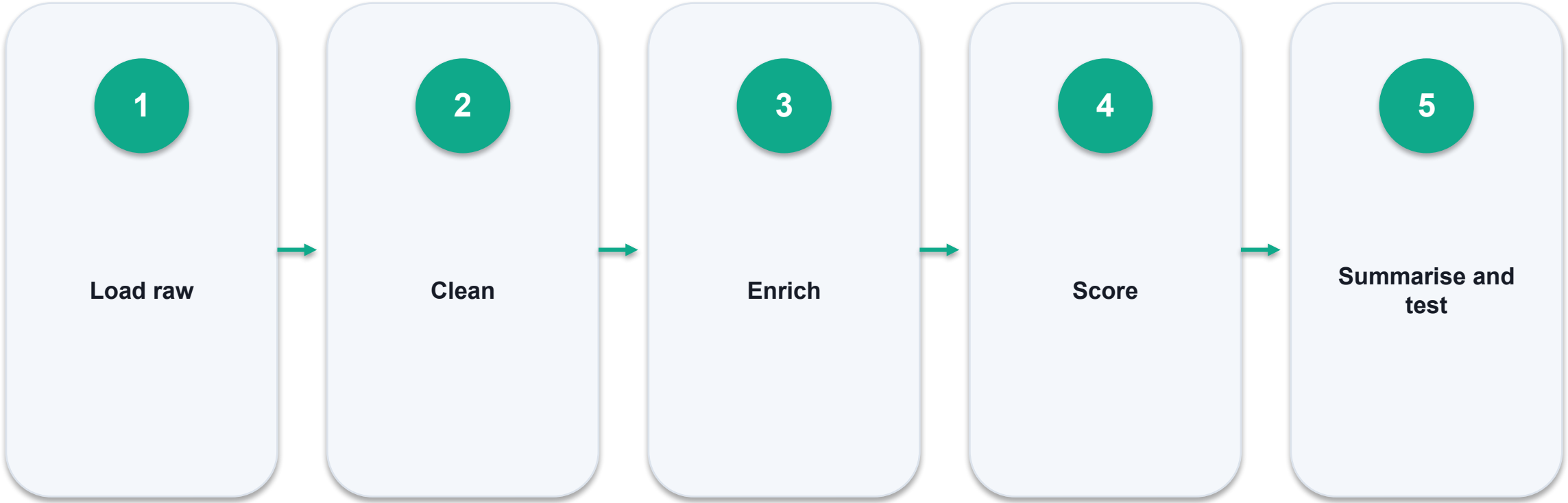
A chain of named clean, enrich, score and summarise functions connected with pipe().

CONTROL **Make each transformation testable; reconcile row counts and risk bands after every stage.**

Pandas, CSV, unittest · Folder: labs/lab-15-*

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Build a Clean-to-Risk Pipeline with pipe — Analytical Workflow



The static slide contains the full sequence; trainers may reveal stages progressively.

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LAB 15 · VERIFY

Build a Clean-to-Risk Pipeline with pipe — Evidence & Acceptance

ACCEPTANCE TEST

Each stage can be tested alone, the raw DataFrame is unchanged, and the final risk-band counts reconcile to total rows.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC 07 · HANDS-ON · LAB 16

Analyse Statistics, Correlation and Rolling Risk

FINANCE SCENARIO

A portfolio risk analyst must profile returns, compare bank-stock relationships and track changing volatility over time.

YOU WILL PRODUCE

A statistics table, correlation matrix, rolling-volatility series and drawdown chart.

TOOLS & EVIDENCE

Pandas, NumPy, Matplotlib, CSV

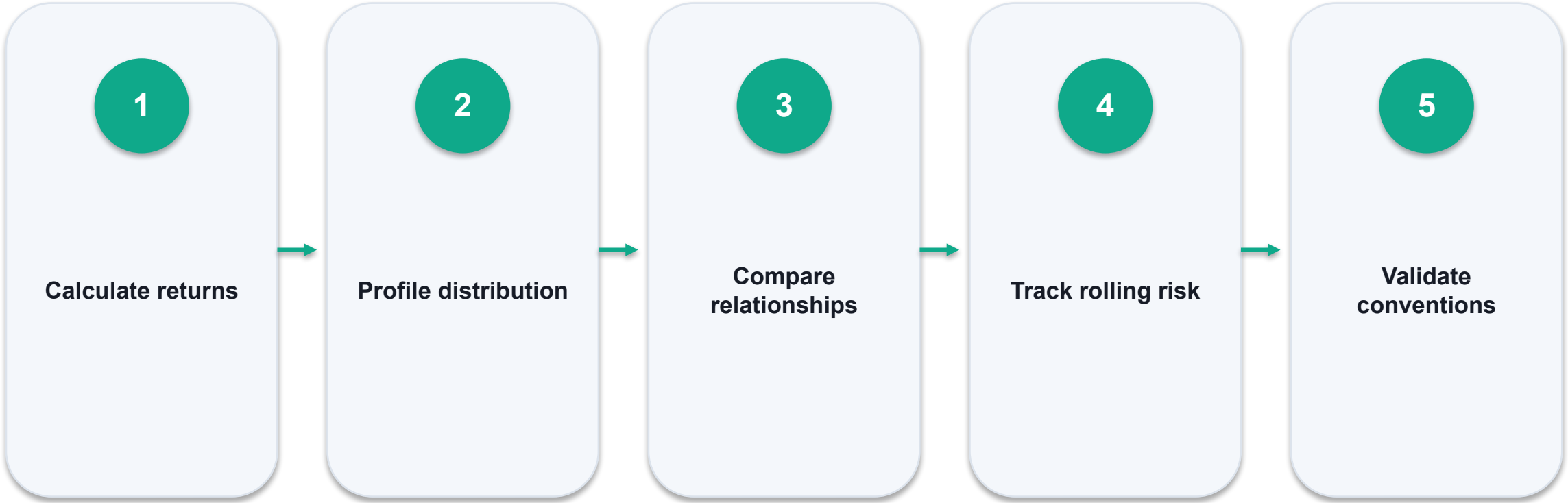
ALIGNMENT

LO7

Folder: labs/lab-16-*

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Analyse Statistics, Correlation and Rolling Risk — Analytical Workflow



The static slide contains the full sequence; trainers may reveal stages progressively.

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LAB 16 · VERIFY

Analyse Statistics, Correlation and Rolling Risk — Evidence & Acceptance

ACCEPTANCE TEST

Correlation is symmetric with 1 on the diagonal, rolling volatility has the expected warm-up NaNs, and maximum drawdown is non-positive.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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Capstone: Produce an Executive Finance Analytics Pack



FINANCE SCENARIO

The CFO requests a concise, reproducible evidence pack combining data quality, portfolio exposure, returns, risk and recommendations.

YOU WILL PRODUCE

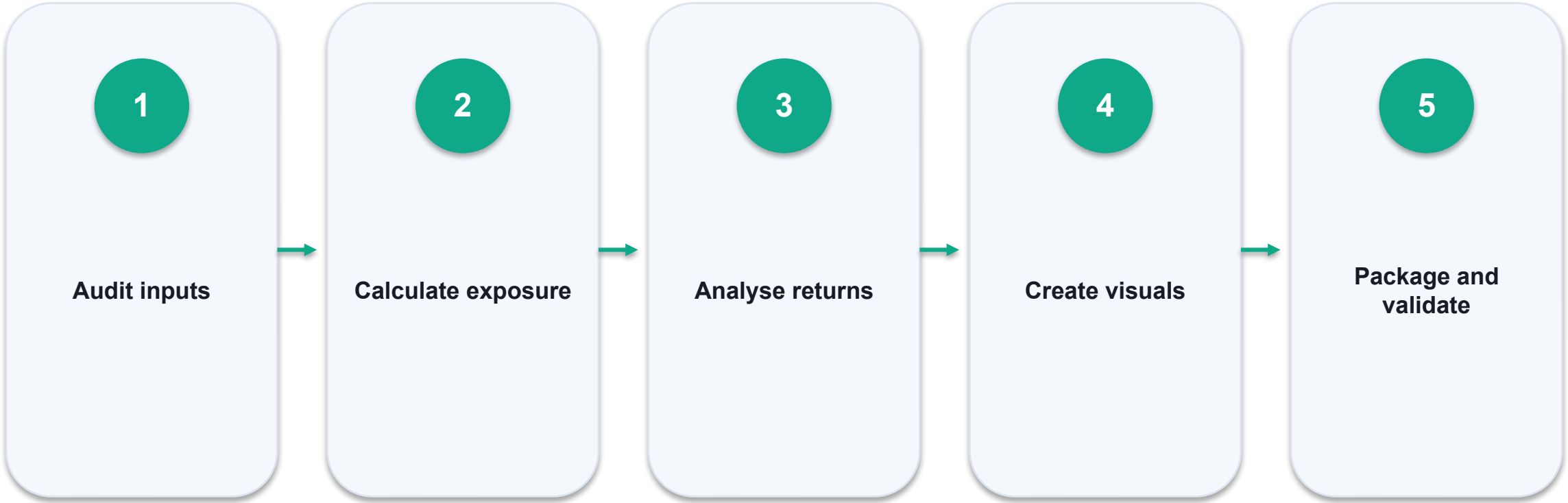
A self-contained HTML report, charts, curated CSVs, validation log and executive summary.

CONTROL **Trace every executive conclusion to supplied data, validated calculations and reproducible outputs.**

Python, Pandas, Matplotlib, HTML · Folder: labs/lab-17-*

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Capstone: Produce an Executive Finance Analytics Pack — Analytical Workflow



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LAB 17 · VERIFY

Capstone: Produce an Executive Finance Analytics Pack — Evidence & Acceptance

ACCEPTANCE TEST

All required files exist, totals reconcile, charts have labels, the HTML report opens locally, and validate_outputs.py prints CAPSTONE PASS.

CAPTURE

1. Executed output
2. Validation result
3. Chart or curated file
4. Short interpretation

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TOPIC RECAP

Recap — Analyze Finance Data

1

Concept

Run a function across rows or columns when vectorised operations are not practical.

2

Method

apply is flexible, but not automatically fast

3

Application

Lab 15, Lab 16, Lab 17

4

Evidence

Validate outputs and explain the finance meaning.

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WRAP-UP

Course Summary & Next Steps

From requirement to explainable finance analytics

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LEARNING OUTCOMES

What You Can Now Do

1

LO1

Translate finance business requirements into clear Python programming objectives and configure a reproducible development environment.

2

LO2

Use Python data types and operators to represent, calculate and present financial information accurately.

3

LO3

Organise finance logic with conditional statements, loops and comprehensions to solve repeatable analytical problems.

4

LO4

Create reusable functions and lambda-based transformations for finance business use cases.

5

LO5

Import, inspect, filter, slice and clean finance data with Pandas Series and DataFrames.

6

LO6

Join, aggregate, test and visualise finance data using groupby, pivot tables and appropriate charts.

7

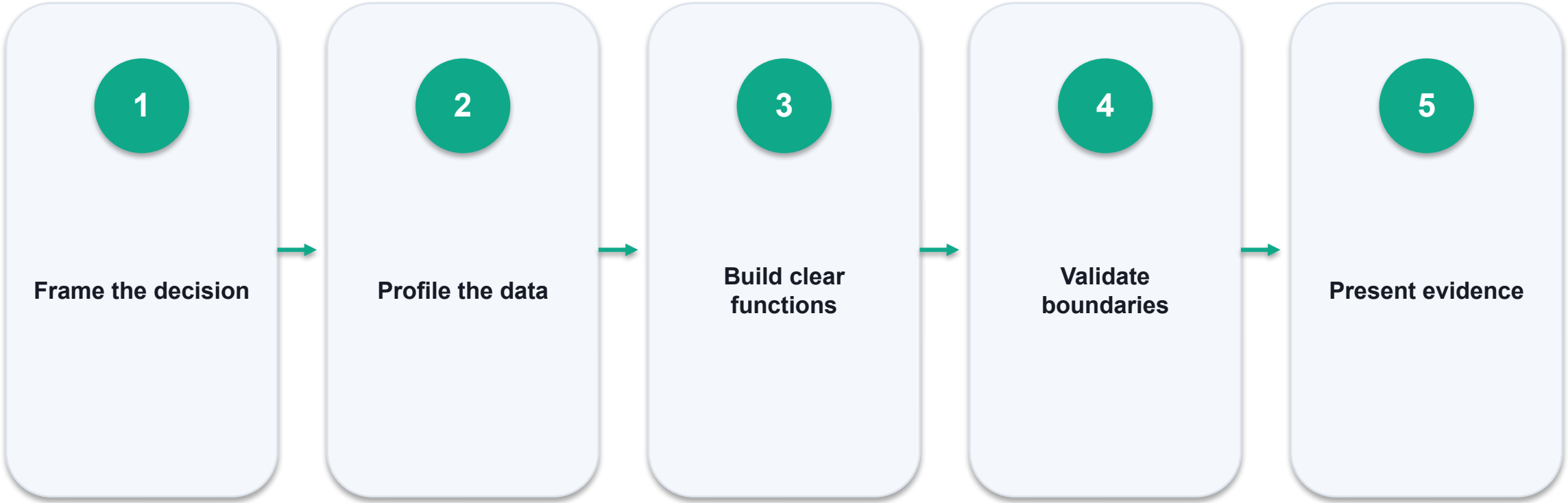
LO7

Improve analytical code with apply and pipe, use descriptive statistics, and analyse changes in financial data over time.

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TRANSFER TO WORK

A Repeatable Standard for Your Next Project



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COURSE PORTAL · LMS/TMS

Download Course Material

https://lms-tms.tertiaryinfotech.com/

1 · SIGN IN — Use the course invitation or trainer-provided credentials.

2 · OPEN COURSEWARE — Choose the current Python Programming for Finance course.

3 · DOWNLOAD — Save the slides, Learner Guide and lab folders locally.

4 · ASSESS — Open and submit the learner-facing papers only when directed.

COURSE PACKAGE

- Visual course slides
- Detailed Learner Guide
- 17 self-contained lab folders
- Starter code and offline data
- Assessment papers on the LMS

Open LMS/TMS

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WRAP-UP

Assessment Reminder

1

WA · 60 min

8 open-ended knowledge questions mapped to K1–K8.

2

PP · 60 min

5 practical tasks mapped to A1–A9 and completed individually.

3

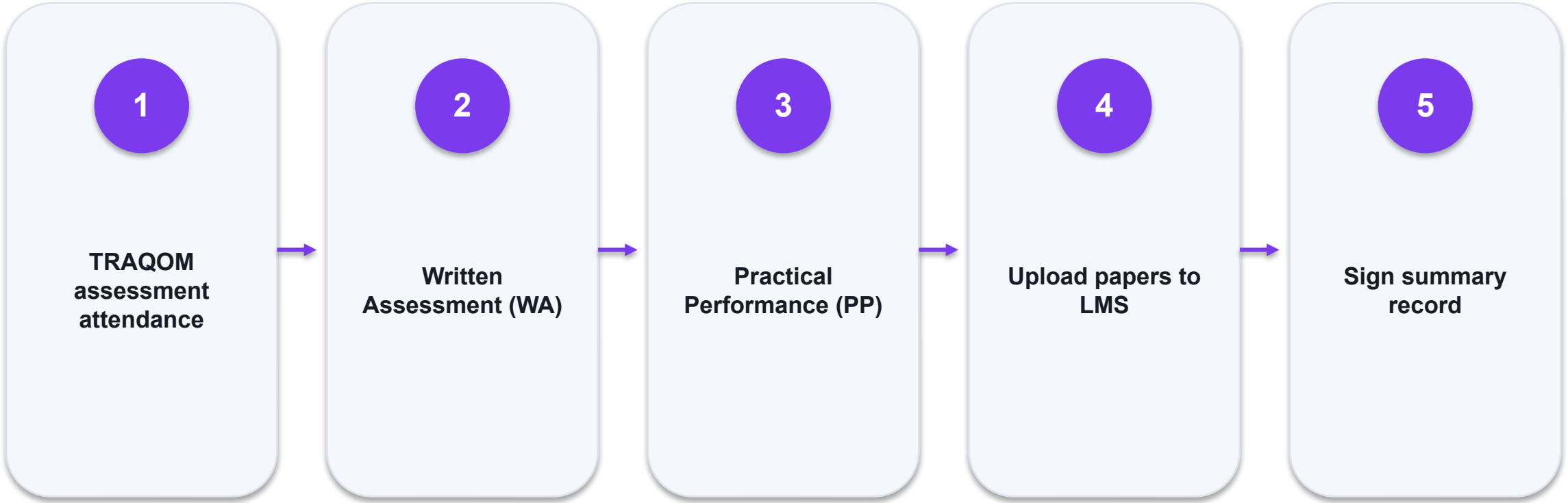
Submission

Upload the two question papers on the LMS and sign the summary record.

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ON ASSESSMENT DAY

Assessment Flow



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Digital Attendance (Mandatory)

1

Assessment window

Scan the separate assessment attendance QR code.

2

Confirm

Check that the submission was accepted.

3

Then assess

Begin only when instructed by the assessor.

4

Evidence

Complete the Assessment Summary Record after both instruments.

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WELL DONE

Thank You!

You can now build, test and explain Python analyses for realistic finance decisions.